DEGRADX: A DATASET-CALIBRATED SYNTHETIC BENCHMARK FOR ATTRIBUTION EVALUATION ON DEGRADING SYSTEMS *

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ABSTRACT

Attribution methods are increasingly used to justify models that estimate the remaining useful life (RUL) of degrading assets, but on measured data the contribution of each observation to an estimate is unknown, so whether an attribution is faithful cannot be checked. We present DEGRADX, a benchmark whose generated multivariate sequences carry a regression target written as an additive function of a nonstationary mean component and of localized patterns, so that the attribution ground truth is known exactly at every observation. The generator is fitted to public battery-degradation datasets, three of which are evaluated, and each evaluated profile is compared with the dataset it stands for. A model trained on the HUST profile predicts the RUL of held-out measured cells with 1.17 [0.94, 1.34] times the error of a model trained on the measured cells; the MATR and FORM profiles keep gaps of 1.49 and 1.54, which in FORM lie in the channels other than capacity, mostly in their levels early in a cell's life. The target is learnable and trained models use every channel it covers, but localized patterns at the amplitude measured in FORM are not shown to be learned. Rank agreement with the ground truth misses smoothed and partly shifted attribution maps, which a temporal profile error introduced here registers. On a reference model whose exact attribution is known, Integrated Gradients, feature occlusion and TimeSHAP recover it; on trained models, ten equally accurate models spread further than the three methods differ in every score of every profile except channel allocation in MATR. The benchmark therefore adjudicates attribution methods on the reference model and reports, with every score, how closely its sequences match the measured process.

***K*eywords** Explainable artificial intelligence · Feature attribution · Benchmarking · Time-series regression · Prognostics and health management · Synthetic data generation

1 Introduction

Industrial assets degrade in service: bearings wear, rotating machinery loses efficiency, and battery cells lose capacity. Condition monitoring records this progression, directly or indirectly, in sensor time series, and maintenance policies that act on it are economically consequential. A survey of United States manufacturers estimates the total annual cost of inadequate maintenance at $222.0 billion and reports that respondents relying more heavily on predictive than

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on preventive maintenance experienced 18.5 % less unplanned downtime and 87.3 % fewer defects [1]. Data-driven models support the tasks on which such policies depend: detecting incipient faults, estimating the current health state, and predicting the remaining useful life (RUL). Because the resulting decisions are consequential, a numerical estimate is seldom sufficient on its own. Before a model is relied upon, engineers need evidence that it responds to the physical signature of degradation rather than to the incidental structure introduced by the acquisition process or by the composition of the training set, and an estimate that triggers an intervention must be justifiable to those who act on it. Post-hoc attribution is by far the most frequently applied method across manufacturing and industrial cyber-physical systems, as it can be applied to a model that is already trained and operational without altering it [2].

Attribution methods assign a value to each element of an input sequence, representing its contribution to the model's output. Whether such a method is faithful (that is, whether the returned values reflect the model's actual dependence on the input) is a separate question from predictive accuracy: a model may estimate its target well while the attribution computed for it misidentifies which inputs drove the estimate. Evaluation of faithfulness requires attribution ground truth specifying the actual contribution of each input element, and on measured data, no such ground truth exists. Validation therefore proceeds by perturbing input elements in order of estimated importance and measuring the effect on model performance. However, the reliability of this procedure is contested at several levels. Occluding elements skew the input distribution, so the model is queried on samples unlike those it was trained on, which is among the drawbacks that motivated dedicated evaluation frameworks for time series [3]. Still, such frameworks have themselves since been shown to contain significant weaknesses [4]. The metric commonly used in region perturbation, the Area Under Perturbation Curve (AUPC), was shown to be inadequate on its own for estimating faithfulness and can support incorrect conclusions, revealing neither how well relevant and irrelevant elements are separated nor how consistently, with results depending materially on the perturbation method applied and region size selection [5]. Under controlled conditions with known relevant elements, the disagreement is systematic, with perturbation-based and ground-truth-based metrics ranking the same attributions in opposite orders [6]. Such measures can therefore establish that two methods differ, but not which of them is closer to the model's actual dependence structure.

The established response is to construct data for which the attribution ground truth is known. This design underlies the synthetic evaluation benchmarks used throughout the time-series explainability literature [7, 8, 9]. However, degradation data differs from the processes these benchmarks generate in three respects. *First*, the target is continuous rather than categorical. *Second*, the signal exhibits sustained mean-nonstationarity, and this component carries the larger part of the target-relevant information. *Third*, the mean component contributes to every observation rather than to a few localized patterns alone, so relevance is graded rather than binary and is not exhausted by a set of positions. Established generators are built on the opposite premise: class-discriminating patterns are planted at known positions, and those positions constitute the attribution ground truth—an approach now standard enough to have its own package [10]. Neither the data nor this form of attribution ground truth transfers to prognostics, where attribution evaluation is therefore confined to functionally grounded indicators of the kind examined above [11].

This work presents DEGRADX, a benchmark for evaluating attribution on multivariate time series whose regression target depends on a sustained nonstationary mean—the regime characteristic of degrading systems. It consists of a generator, a set of calibrated configurations, and an evaluation protocol. The generator produces multivariate sequences by adding a nonstationary mean component, localized patterns inserted at known positions, and noise. The target variable is an explicit additive function of the first two terms, so the contribution of every observation is known by construction rather than estimated. This yields attribution ground truth of two kinds: a dense graded field from the mean component, which contributes at every position, and a sparse set of positions carrying the inserted patterns. The protocol measures each on its own terms: rank agreement, a channel allocation error, and a temporal profile error against the graded field, and precision-recall retrieval metrics standard in the literature [12, 9] against the sparse set. Retrieval is scored on each window paired with its counterpart without inserted patterns, since on a single map the graded field outweighs the patterns and ranking the cells of the attribution ground truth itself retrieves them barely above chance. Its values are therefore not comparable with retrieval scores reported on existing benchmarks, and it is computed only where a profile carries inserted patterns, which of the profiles evaluated here only FORM does; there a trained model is not shown to learn the pattern term, so its retrieval scores cannot be read, and they are reported for the reference model and for degraded ground-truth maps alone. The generator parameters are fitted to public degradation datasets **MATR** [13], **HUST** [14], and **FORM**, the formation dataset of [15], whose cells differ from those of MATR and HUST in chemistry, format and laboratory, each fitted configuration being called a *profile*; a fourth dataset, **ISU-ILCC** [16], was fitted as a candidate and is reported but not evaluated, since the levels set by its test conditions carry more than the declared half of the variance of its target (Section 3.4). Synthetic evaluation invites the objection that conclusions drawn on generated data need not hold on measured data, and existing benchmarks acknowledge this gap rather than measure it, treating realism and ground-truth availability as a trade-off [17]. We instead report it: every evaluated profile is validated against the dataset it was fitted to, by distributional comparison and by the "Train on Synthetic, Test on

Real" (TSTR) framework, so that each score is accompanied by evidence of how closely the profile reproduces the process it stands for. We therefore investigate three properties of DEGRADX.

- **RQ1:** How closely does each profile reproduce the dataset it was fitted to, in distribution and in task performance, and which properties resist calibration?
- **RQ2:** Is the resulting attribution ground truth usable, that is, learnable by a trained model, and does such a model use each channel that enters the target?
- **RQ3:** Do the scores discriminate between better and worse attribution maps, and which failures of a map does each of them register?

The proposed benchmark is intended for any setting in which attribution is applied to a regression target under sustained mean-nonstationarity. Within prognostics, it supplies the attribution ground truth that evaluation currently lacks, allowing a method to be assessed before it is applied to a model. Beyond it, the benchmark extends the evaluation surface of perturbation-based explainers, which are at present assessed almost exclusively on classification tasks over stationary or state-switching processes. The generator, fitted parameters, seeds, trained models, ground-truth files, and scoring code are publicly released [2].

2 Related Work

The construction of synthetic data with known attribution ground truth is well developed for classification. Ismail *et al.* [7] built 70 datasets from seven base processes, including Gaussian noise, autoregressive variants, a Gaussian-process mixture, harmonic functions, and NARMA, marking informative features by an offset applied to one class, and evaluated saliency methods across architectures. Tonekaboni *et al.* [8] introduced the "feature importance in time" (FIT) framework, which was subsequently reused across the perturbation-based explainability literature [12, 18, 19]. Queen *et al.* [9] designed four synthetic datasets on a comparable base process and evaluated them together with four measured ones. Packaged environments have followed, covering perturbation-, gradient-, and example-based explainers [20], distributing pretrained models with precomputed explanations [21], and supplying the planting procedure itself as reusable software [10]. In every case, the attribution ground truth is defined by class-discriminating structure, an offset applied to one class, or a transient that determines the label. The same construction has more recently been turned on the evaluation procedure itself, using planted data to establish how perturbation-based and ground-truth-based scores diverge [6], and the authors of the most recent of these resources state that it is limited to classification, listing forecasting among the extensions left to future work [21].

A second group of benchmarks provides ground truth that describes the generating process rather than an individual prediction and therefore cannot be used to score an attribution map. Barić *et al.* [22] specified synthetic datasets whose generating process describes interactions between multiple series at increasing complexity and reported that models with satisfactory predictive performance frequently yield incorrect interpretations—an observation consistent with the separation between accuracy and faithfulness set out above. Their ground truth, however, is the interaction structure itself: which source series and which lags drive the target. TimeGraph [23] identifies the same neglect of nonstationarity that motivates this work, noting that trends, seasonality, irregular sampling, and unobserved confounders are generally absent from synthetic benchmarks, and supplies datasets that incorporate them. Its dynamics are therefore appropriate, but its ground truth is again structural—a causal graph over series and lags.

Attribution ground truth for regression is recent and takes two forms. *First*, series carrying known attribution ground truth are generated and then added to measured datasets so that the resulting semi-synthetic data retains real-world distributions. Kamarthi *et al.* [24] apply this to hierarchical demand forecasting, adding generated series to four measured datasets and evaluating attribution from the resulting variation in the outputs. Attribution is nonetheless scored against the attribution ground truth of the generated series, which are built on a noise background and are surrounded but not altered by the measured datasets. Therefore, the signal being scored exhibits no sustained mean-nonstationarity. *Second*, the attribution ground truth is computed rather than planted. When the target variable is not an additive function of its inputs, it cannot be decomposed into one contribution per input, and an attribution method must instead be applied to the generating process itself, whose output is then taken as the attribution ground truth. This has been done for forecasting series carrying seasonal structure and external inputs [25]. The quantity recovered is the importance of each input variable across the dataset rather than the contribution of individual observations. Where no attribution ground truth is available, evaluation falls back on functionally grounded indicators, that is, criteria computed from model behavior alone. A recent survey of explainable AI for industrial RUL prediction builds its entire evaluation framework from indicators of this kind [11], and the same practice is adopted by benchmark resources whose measured

[2] `[PLACEHOLDER: repository and DOI]`

datasets carry no relevance annotation [21]. The reliability of this practice is disputed. Under controlled conditions with known relevant elements, indicators of this kind and ground-truth-based measures rank the same attributions in opposite orders [6], and work on benchmark fidelity attributes part of the disagreement to confounding factors in the evaluation setup itself [26].

A separate group of works produces realistic synthetic sequences that closely resemble measured data without providing any attribution ground truth. Diffusion-based [27], generative-adversarial [28], and hybrid [29] models generate capacity-fade curves fitted to public datasets and report that data of this kind can replace a substantial fraction of laboratory measurements in prediction tasks. In each case, the generating function is learned rather than written down, so no explicit mapping from individual observations to the target variable exists, and their contributions cannot be recovered. These models therefore occupy the position complementary to the measured data, but not as a ground-truth benchmark for attribution methods. This literature has, however, established how the distance between generated and measured data is estimated. Esteban *et al.* [30] introduced the TSTR procedure, training a model on generated data and reporting its error on measured data, which tests whether the dependence a model has to learn was preserved rather than only the appearance of the sequences. Yoon *et al.* [31] paired it with the classification error of a discriminator trained to separate the two, and Jeha *et al.* [32] added a Fréchet-style distance computed in a learned representation space. Measures of this kind are reported by the degradation generators above [27], and by *none* of the attribution benchmarks of the preceding paragraphs. The authors concede the gap between generated and measured data and treat realism and ground-truth availability as a trade-off, given that the generation process for measured data is unknown [17].

Across the overviewed body of work, the properties DEGRADX requires are available separately and never together. The synthetic benchmarks supply exact attribution ground truth, but for a categorical target and in the form of planted positions, which a component present at every observation cannot be reduced to. The regression benchmarks provide either attribution ground truth for a signal with no sustained nonstationarity or ground truth describing the generating process rather than individual observations. Where neither is available, the evaluation proceeds with no reference at all. The generators of degradation data supply both the signal and the measured correspondence to it, but their generating function is learned, and the contribution of an observation cannot be read off it. DEGRADX occupies the position that remains. It supplies attribution ground truth defined at every observation for a continuous target under sustained mean-nonstationarity, and it applies the measures of the generative literature to an attribution benchmark: each profile is compared with the dataset it was fitted to in distribution and under TSTR by whether a model trained on the profile predicts on that dataset. The second is the more demanding of the two, since it tests the dependence between the sequences and the target variable rather than the sequences alone. How closely a profile matches on both is reported with every score it produces. The match bears on the sequences on which attribution methods are tested and on their relation to the RUL, not on the attribution ground truth, which follows from a target written down for the benchmark and has no measured counterpart: the closer the match, the stronger the case that methods are scored on sequences like those of the measured process—a case that existing benchmarks, by the trade-off conceded above, cannot make.

3 Methodology

DEGRADX consists of a generator, a fitting procedure that adapts the generator to a measured dataset, and an evaluation protocol. This section describes them in that order. Section 3.1 defines the generated sequences, Section 3.2 the targets and the attribution ground truth derived from them, Section 3.3 the fitting procedure, Section 3.4 the datasets it is applied to, and Section 3.5 the measurements that answer the three research questions. Figure 1 shows how the parts connect.

3.1 Generator

A generated unit consists of a multivariate sequence together with the targets and the attribution ground truth derived from it in Section 3.2. The sequence is $x \in \mathbb{R}^{T \times C}$, where $t \in \{1, \ldots, T\}$ indexes positions and $c \in \{1, \ldots, C\}$ indexes channels. In the datasets used here one position corresponds to one charge–discharge cycle, and T is the position at which the unit reaches its end of life (EOL), defined in Equation 3 below. Each channel is the sum of three terms,

$$x_{t,c} \;=\; m_{t,c} \;+\; \sum_{k} p_{k,t,c} \;+\; \varepsilon_{t,c}, \tag{1}$$

a mean component m, a set of inserted patterns $p_{k,t,c}$, and noise ε. Figure 2 shows the three terms and their sum for one channel. This composition, a base process into which patterns are inserted at recorded positions, is the design used by established synthetic benchmarks [8, 9]. What differs is the role of the first term. There the base process is background, and the attribution ground truth does not depend on it; here the mean component carries the degradation trajectory and enters the target variable, which is what makes it graded rather than confined to the inserted patterns.

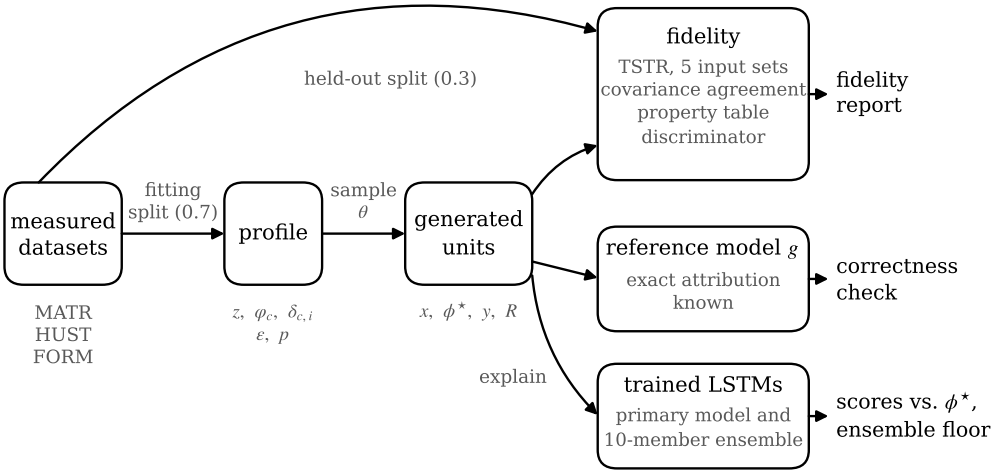


Figure 1: Structure of the benchmark. Generator parameters are estimated from the fitting split of a measured dataset (MATR, HUST or FORM; 0.7 of its units, the remaining 0.3 held out), giving a profile: the degradation state z, the channel mappings φ_c shared by all units with a per-unit offset $\delta_{c,i}$, the noise ε, correlated across channels, and the inserted patterns p (Section 3.1). Drawing θ per unit generates units, each carrying the sequence x, the attribution ground truth $\phi^\star$, the decomposable target y, and the transfer target R (Section 3.2). Attribution maps are scored on generated units only, since $\phi^\star$ is known nowhere else: on the reference model g, whose exact attribution is known, as a correctness check of each method, and on trained models, for the primary model and for an ensemble of ten equally accurate models whose spread is the identifiability floor (Section 3.6). The profile is separately compared against the held-out split, which took no part in fitting it, by the transfer ratio for each of five input sets, the covariance agreement, the property table and a discriminator (Section 3.5).

3.1.1 Mean Component and Degradation State

Degradation is one underlying process observed through several measured quantities, and those quantities do not follow the same shape: e.g., capacity falls while internal resistance and charge time rise. Fitting one parametric curve to every channel would impose a shape that only some of them have, while fitting an independent curve to each channel would instead leave the relationship between channels unconstrained. The generator therefore separates *how far a unit has degraded* from *what that degree of degradation looks like in a given channel*. The first is a scalar *degradation state* z_t, the second a mapping φ_c per channel, to which each unit i adds a constant offset $\delta_{c,i}$,

$$m_{t,c} \;=\; \varphi_c(z_t) \;+\; \delta_{c,i}. \tag{2}$$

The mappings φ_c are smooth, are shared across all units of a profile, and are fitted from the measured data as described in Section 3.3. The offset $\delta_{c,i}$ shifts unit i in channel c by the same amount at every position, for reasons other than degradation, such as the protocol under which the unit was cycled; the offsets of a profile have median zero over its units, so that φ_c describes the median unit. The capacity channel carries no offset: its mean component follows from the definition of the degradation state below, taken with each unit's own early-life reference, so that units start from different capacities and meet at the EOL threshold. Units therefore differ in the pace at which they age and in their levels, but not in what a given degree of ageing looks like: two units that have reached the same degradation state show the same expected values in every channel other than capacity, up to their offsets, at whatever positions they reached it. Figure 3 illustrates this for two units with different lifetimes. The differences in level between units are part of the mean component and therefore enter the decomposable target of Section 3.2. Their share of the variance of that target is reported for every profile; a profile in which it exceeds one half is not taken through the evaluation protocol of Section 3.5, since its target would then record mainly the level at which a unit sits rather than how far it has degraded. Because every channel is a function of the same degradation state, a channel that does not enter the target still carries that state, and whether a model relies on it cannot show whether the generated data leaks the target. Each profile therefore also contains two null channels with no dependence on the degradation state: one with a constant mapping plus noise, and one whose values are drawn from the measured values of a channel with unit identity and order removed by permutation. Both have zero weight in the target, and they are the channels on which leakage is tested (Section 3.5).

Definition of the degradation state and of EOL. Let q_t denote the capacity channel, smoothed by a declared filter, let q_{nom} be the nominal capacity documented for the dataset, and let q_1 be the maximum of q_t over a declared number k

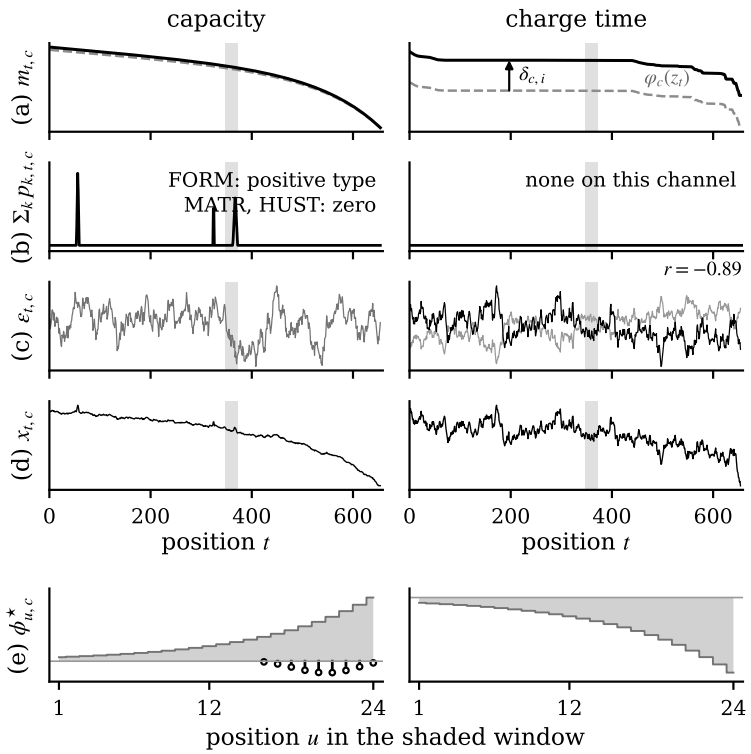


Figure 2: Two channels, capacity (left) and charge time (right), of one generated FORM unit: (a) mean component, with its shared part $\varphi_c(z_t)$ dashed; on charge time the two differ by the unit's offset $\delta_{c,i}$, (b) inserted patterns, here positive capacity recoveries, the one type FORM enables (in the measured cells they follow diagnostic blocks; the generator inserts them at uniformly drawn positions), and zero in MATR and HUST, (c) noise, standardised; the noise of mean discharge voltage (grey) is drawn with that of charge time at their fitted within-unit correlation, $r = -0.89$, (d) their sum, which is what a model receives, and (e) the resulting attribution ground truth of the shaded window of $L = 24$ positions under recency weighting, a dense graded field (shaded) from the mean component and a sparse set of positions (stems) from the inserted patterns. Vertical scales are arbitrary.

of initial positions, attained at position t_1. EOL is reached when capacity has fallen to a fraction ρ of nominal capacity, and the degradation state measures how much of the fall from q_1 to that threshold has been consumed,

$$z_t \;=\; \frac{q_1 - q_t}{q_1 - \rho\, q_{\mathrm{nom}}}, \qquad T \;=\; \min\{\, t \geq t_1 : z_t \geq 1 \,\}. \tag{3}$$

The degradation state runs from 0 at the early-life reference to 1 at EOL. Because both endpoints are fixed in advance, $\rho\, q_{\mathrm{nom}}$ by declaration and q_1 by observation, z_t is a transform of the current capacity alone. It is therefore computable from data available up to t, once the first k positions have been observed, and T is a consequence of the trajectory rather than an input to it. The early-life reference is a maximum over the first positions rather than the first value because measured capacity can rise over early cycles before it falls, and a first-position reference would make the degradation state negative over that interval; how often the two references differ is reported with the property audit of Section 3.4. The more obvious alternative—rescaling each unit's trajectory to the unit interval over its observed length—would make z_t a function of t/T. In this case, the degradation state, and through Equation 2 every channel, would encode the unit's total lifetime at every position. Since total lifetime is what the transfer target of Section 3.2 asks a model to predict, the generated data would exhibit target leakage, and a model trained on it would depend on information that is unavailable at prediction time, a failure mode reported for RUL pipelines [33].

Measured datasets do not agree on ρ. The common convention is a fall to 80% of nominal capacity, but the experiments underlying some datasets were stopped at a fade of 20 to 30%, and individual units within a dataset were carried to different final capacities. Adopting each dataset's own convention would define z_t against a different endpoint in each profile and would render the parameter distributions fitted from them mutually incomparable. We therefore apply a single declared ρ across all profiles, taken against the nominal capacity each dataset documents rather than against a unit's own initial capacity. The experiments stop cycling at thresholds stated relative to nominal capacity, so a fraction

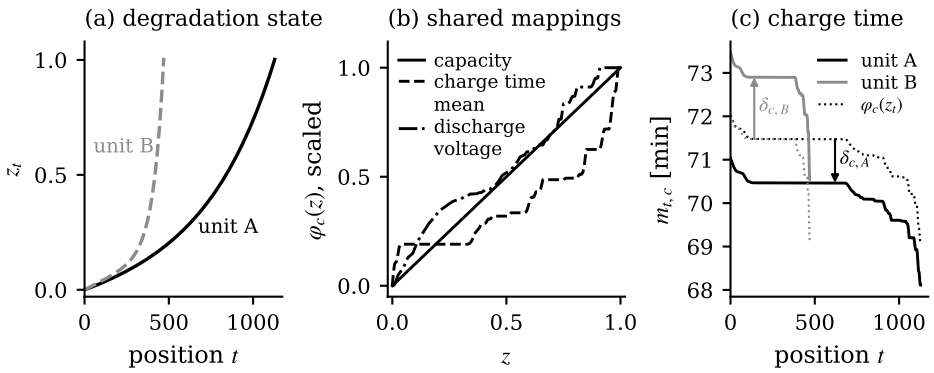


Figure 3: Construction of the mean component, on the FORM profile. Two units advance through the degradation state at different paces and reach EOL at different positions (a). The same fitted mappings (b), scaled to their change from the pristine level to EOL, are applied to both. In charge time (c), each unit's mean component is the shared mapping at its own state (dotted) shifted by the unit's offset $\delta_{c,i}$ (solid), so the two units differ in timing through z_t and in level through $\delta_{c,i}$; capacity carries no offset.

taken against a higher initial capacity would place EOL below the point at which cycling stopped and would exclude units for a reason unrelated to how they degrade. A unit whose early-life reference lies within a declared margin of the threshold is excluded, since the permitted fall is then too small to define the degradation state. Some datasets were recorded only until capacity reached the stopping threshold, so a measured unit whose last recorded capacity lies within a declared tolerance of the threshold is taken to reach EOL at its last position, which is where the experiment itself located it; generated units reach EOL where Equation 3 places it and are stored only up to that position, so the same tolerance applies to them wherever their properties are estimated with the procedure of Section 3.3. We report the number of units that never reach the threshold, which are excluded from the quantities that depend on EOL (the parameter distribution, the length distribution, and the transfer target) and retained for those that do not (the channel mappings, the noise, and the inserted patterns), since their degradation state is defined, and report the sensitivity of the fitted parameter distributions to ρ across the range these conventions span, including the definition that takes ρ against the first-position capacity.

Shape of the degradation state. The trajectory q_t, and hence z_t, follows a parametric curve with a low-dimensional parameter vector θ drawn per unit from a distribution estimated from measured units. The functional form is not fixed in advance. Several established empirical fade models are fitted to each dataset and the one that fits best is selected, so that the shape carried by a profile is decided by the data it stands for rather than by us; Section 3.3 gives the candidates and the selection rule.

3.1.2 Inserted Patterns and Noise

Inserted patterns. Measured degradation trajectories are not smooth. Alongside the global degradation trend, capacity sequences carry local fluctuations, and the established works separates the two, decomposing the trajectory into a degradation trend, a regeneration component, and a residual fluctuation component [34]. The positive fluctuations are *capacity regeneration*: a temporary recovery of measured capacity that follows a rest period between cycles and persists over the cycles after it [35, 36]. Negative fluctuations follow from the conditions under which a cycle is measured rather than from the state of health of the cell; ambient temperature is the documented instance, appearing as chamber temperature excursions in public cycling datasets [13]. In both cases the deviation is treated as separate from the degradation trend, which is estimated and extrapolated in its own right [34, 35], and not as a change in how far the unit has degraded. In this work, these fluctuations are incorporated as profiles without them would be smoother than the dataset they stand for, and the fidelity measurements would record the difference. They also supply the localized part of the attribution ground truth, which is the part the retrieval metrics reported on existing benchmarks are defined on. The generator represents these effects as inserted patterns $p_{k,t,c}$, each described by a position, a signed amplitude, and a duration. Because they act on the observation and not on the degradation state, they do not alter z_t, the position of EOL, or the transfer target. Section 3.3 states how they are detected in measured data and what is estimated from them. A profile enables only the inserted pattern types its dataset exhibits.

Noise. Noise is added per channel with variance and autocorrelation set during fitting, and has zero mean, $\mathbb{E}[\varepsilon_{t,c}] = 0$. Because retrieval scores depend on the signal-to-noise ratio, a generator cleaner than the measured data would report

every attribution method as more accurate than it is. Within a channel the noise is a first-order autoregressive process. Across channels it is not independent: the residuals of measured channels can move together within a unit, and the cross-channel covariance is one of the fidelity measures of Section 3.5, so the innovations of the channels of a unit are drawn jointly. One correlation matrix of the innovations is used per profile, chosen so that, at the autocorrelations fitted for the profile, the stationary correlation of the noise between two channels equals the within-unit correlation of their residuals after the offsets are removed (Section 3.3); each channel of a unit keeps its own variance and autocorrelation. A matrix that is not positive definite is replaced by the nearest one that is, and the change is reported. The null channels stay independent of every other channel.

3.2 Targets and Attribution Ground Truth

Two quantities have to be distinguished. A *target* is a value a model is trained to predict from a window of the sequence. The *attribution ground truth* is the contribution that each position and channel of that window makes to the target; it is the quantity an attribution method is asked to recover and against which its output is scored. The generator defines two targets, for two different purposes. The *decomposable target* y is written down as an additive function of the generated components, so that its attribution ground truth follows from its form, and is the target used whenever attributions are scored. The *transfer target* R is the RUL, has a counterpart in every measured dataset, and is used only to test how well a profile reproduces the dataset it was fitted to.

3.2.1 Decomposable Target and Attribution Ground Truth

Let W be a window of L consecutive positions ending at position t, and let u index the positions within it, $u \in \{t - L + 1, \ldots, t\}$. The decomposable target on that window is a weighted sum of the generated components:

$$y \;=\; \sum_{(u,c)\,\in\,W} w_{u,c}\left(m_{u,c} - x_c^0\right) \;+\; \sum_{(u,c)\,\in\,W} w_{u,c} \sum_k p_{k,u,c}. \tag{4}$$

Since $m_{u,c}$ is $\varphi_c(z_u)$ up to the unit's levels (Equation 2), the first term is a fixed function of the degradation state over the window and of those levels: y summarises how far the unit has degraded across those positions, perturbed by whatever transients occurred in them, and it also carries each cell's channel levels through the per-unit offsets. On the generated test windows the offsets carry 0.13, 0.34 and 0.16 of the variance of y in MATR, HUST and FORM under recency weighting, and y falls with the remaining useful life at a Spearman correlation of -0.71, -0.61 and -0.69 (Section 4.2). It is important to note that equation 4 *is not a physical law and is not proposed as a model of how RUL arises from a degradation trajectory*. The decomposable target is never compared against measured data, whereas the sequences and the transfer target are. Because Equation 4 is additive and written down rather than learned, the contribution of every position and channel is known exactly:

$$\phi^\star_{u,c} \;=\; w_{u,c}\left(m_{u,c} - x_c^0\right) \;+\; w_{u,c} \sum_k p_{k,u,c}. \tag{5}$$

The first term of Equation 5 is the graded field and the second the sparse set. Both are measured from the reference point $x_c^0 = \varphi_c(0)$, the pristine level of channel c, which is declared per profile and identical across its units. It is the input at which the decomposable target is zero, so a unit that has not degraded, sits at the median levels of its profile, and carries no inserted pattern has $y = 0$, and an attribution method whose values sum to the difference between the model output on the window and on its baseline recovers $\phi^\star$ only when that baseline is x^0 (Section 3.6), and a target measured from it can be reproduced by a map of the observed window, which a target containing the mean component itself cannot (see the reference model below). There is one weight per position and channel, $w_{u,c} = \kappa_c\,\pi_u$, the product of a channel factor and one of the position profiles below. The channel factor is zero on channels that do not enter the decomposable target and is declared in units of the degradation state on those that do, $\kappa_c = \beta_c/(\varphi_c(1) - \varphi_c(0))$, so that the full change of a channel between the pristine level and EOL contributes a declared share β_c in every profile. Channel roles therefore differ in whether κ_c is zero and whether the channel carries inserted patterns, not in the form of the weight. Every generated window is supplied together with its counterpart in which the inserted patterns are absent, and the two fields are scored on this pair: a method's attribution on the counterpart against the graded field, and the difference between its attributions on the window and on the counterpart against the sparse set, since for $\phi^\star$ these are exactly the two fields. Ranking the cells of a single map would not serve, because the graded field outweighs the patterns on the channel that carries them, so that even $\phi^\star$ would rank the sparse set last. The position profile π_u decides how the relevance of the mean component and of the inserted patterns is distributed across the positions of a window, and therefore what shape a correct attribution map has.

The weights cannot be estimated from measurement: the purpose of y is to be exactly decomposable so that the contribution of every observation follows from its form instead of being estimated; it has no counterpart in a measured

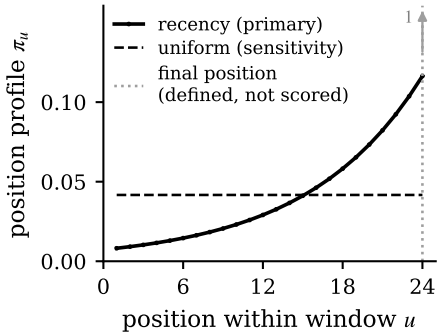


Figure 4: Weighting of positions within a window of $L = 24$ positions. The recency weighting ($h = 6$) used by default keeps every position relevant while giving the graded attribution ground truth an ordering; the alternatives are degenerate in the two ways described in the text: uniform weighting is reported as a sensitivity, and final-position weighting is defined but not scored.

dataset. Fitting the weights would require an external criterion, and both available criteria serve a different purpose. Deriving them from an attribution method applied to a trained model would make the attribution ground truth a function of one of the methods under evaluation. Fitting them so that y approximates the transfer target would make y a second estimate of a quantity the generator already records as a label. It is also indeterminate in the direction that matters: the mean component at adjacent positions within a window is close to collinear, so a fit pins down the aggregate weight far more tightly than its distribution across positions, which is what π exists to specify. Fitted weights would also vary between profiles, making scores incomparable for the same reason ρ is standardized.

Figure 4 shows the three weightings defined here. *Recency* weighting increases smoothly toward the end of the window at a decay rate declared per profile; it keeps every position relevant while giving the field an ordering a method can be scored against. *Uniform* weighting makes every position equally relevant; it is the limit in which no ordering exists to recover. *Final-position* weighting places all weight on the last position; it is the opposite limit, in which both the graded field and the sparse set collapse into the last position of the window, so that an inserted pattern contributes only where it covers that position, and it matches an explained model that reads only the last observation of a window. Recency weighting is primary, and uniform weighting is a sensitivity readout. Reporting both therefore measures how far a result depends on where relevance is placed rather than on which method produced it. The attribution methods under evaluation are scored against $\phi^\star$ under both weightings, and we report whether the score orderings place the methods in the same sequence. Final-position weighting is defined but not scored. Under it $\phi^\star$ is zero at every position but the last, so almost all the cells over which rank agreement is computed are tied in $\phi^\star$. The temporal profile error of Section 3.5 would be defined there, but it is not evaluated either, and how it behaves when relevance is confined to one position is left open.

Reference model. Attribution is evaluated on a trained model, so a low score may reflect error introduced by the attribution method or error in the model itself. To separate the two, the benchmark also provides a *reference model*: the affine map $g(x) = \sum_{(u,c)\in W} w_{u,c}\,(x_{u,c} - x^0_c)$, which applies the weights w to the deviation of the observed window from the reference point. It reproduces the target exactly when noise is absent and otherwise differs from it only through $\sum_{(u,c)\in W} w_{u,c}\,\varepsilon_{u,c}$, whose expectation is zero since $\mathbb{E}[\varepsilon] = 0$. Its intercept does not enter attributions computed from the reference point. Because the map is affine and its weights are known, attributions computed for it can be compared against an analytically available solution, and any remaining error is attributable to the attribution method alone. The reference model addresses the model and not the target: it establishes what an attribution method achieves when the model is exactly right, and says nothing about whether the target is well chosen, which is what the two properties above report.

3.2.2 Transfer Target

The decomposable target has no counterpart in a measured dataset, so a model trained to predict it cannot be evaluated on measured data. The generator therefore records a second label, the RUL at position t,

$$R_t \;=\; T - t, \tag{6}$$

with T as defined in Equation 3. This is the same quantity that measured datasets provide as a RUL label. It is used only for the transfer measurement of Section 3.5 and forms no part of the attribution ground truth.

Two points follow from Equation 6. First, R is not a function of z_t: two units that have reached the same degradation state have different RUL according to how fast they are advancing through it. This is what makes the transfer measurement informative, since a profile whose fitted parameter distribution advances units at the wrong pace will fail it even when the generated sequences match the measured ones in distribution. Second, R depends on the whole trajectory and is therefore a label and never an input. Neither R, nor T, nor any quantity derived from them is available to a model at prediction time, a restriction stated in full in Section 3.3.

3.3 Fitting the Generator to a Measured Dataset

A *profile* is the set of generator parameters obtained from one measured dataset. Fitting uses the training split only, so that the units later used to assess the profile take no part in producing it. The procedure has six steps.

First, the capacity channel is extracted for every unit, readings that no degradation process produces are removed (in any channel, a single position departing from its neighbourhood by more than a declared fraction of its typical value, which the detection rule below could never admit as a pattern; in a channel other than capacity, also a run of up to a declared number of such positions bounded by positions that do not depart, such as a cycle interrupted by a pause together with the disturbed readings after it, which cannot be a pattern since patterns are detected on capacity alone; and readings below half of nominal capacity from which the record later recovers), and the series is smoothed by a Savitzky–Golay filter, and EOL is located by Equation 3. *Second*, candidate parametric families are fitted to each unit's capacity trajectory from the early-life reference to EOL by nonlinear least squares. The families considered are the two-term exponential established for capacity fade [37], a rollover form whose parameters are the initial rate, the transition position, the transition width, and the terminal rate [38], and a power law, which has fewer parameters but cannot express a transition. The family is selected per dataset by cross-validated fit error across units, among the families whose parameters are determined by the data rather than held at a bound of their range in more than a declared share of units, and a family with fewer parameters is preferred within a declared margin of the lowest error; the error of every candidate is reported. The empirical distribution of the fitted parameters over units is the profile's distribution of θ; a generated unit resamples one fitting unit that reaches EOL and takes its θ together with its early-life reference, its offsets, its noise parameters, and its pattern rate, so that the dependence among them across units is kept. *Third*, the mappings φ_c of Equation 2 are fitted as smooth monotone functions of the fitted degradation state, jointly with the offsets $\delta_{c,i}$, and the covariance among channels is recorded; mappings and offsets are fitted by backfitting, which alternates a fit of the mapping to the readings less the offsets with an update of each offset to the median over the unit's positions of its readings less the mapping, centred to median zero over units, until neither changes by more than a declared tolerance or a declared number of iterations is reached; for the capacity channel the mapping is affine by construction, since Equation 3 defines the degradation state as an affine transform of that channel, and it is taken with each unit's own early-life reference. The same affine mapping taken at the median early-life reference of the fitting units is the shared capacity mapping φ_c, which defines the pristine capacity level $x_c^0 = \varphi_c(0)$ and the channel factor κ_c of Section 3.2. The reference point therefore stays identical across units, so the reference model remains a function of the window alone and stays exact, and an undegraded unit at the median levels of its profile still has $y = 0$. *Fourth*, the residuals of the trajectory fit, and in the other channels the residuals about the mapping plus the unit's offset, supply the noise variance and autocorrelation, computed with the positions occupied by detected patterns excluded, so that transients are not counted as noise; their correlation between channels, estimated within each unit and averaged over units, sets the correlation of the noise across channels. *Fifth*, the same residuals supply the inserted patterns, by the detection procedure stated below. *Sixth*, the empirical distribution of unit lengths is recorded.

The distinction between what is measured and what is chosen is reported explicitly, since it determines what the benchmark can claim. Estimated from the measured dataset are the distribution of θ, the choice of family, the mappings φ_c, the per-unit offsets $\delta_{c,i}$ and early-life references, and the covariance among channels, the noise variance and autocorrelation together with the within-unit correlation of the noise between channels, the rate, amplitude, and duration of the inserted patterns, and the length distribution. Declared by design are the EOL fraction ρ, the nominal capacity of each dataset as documented, the number k of positions over which the early-life reference is taken and the margin below which a unit is excluded, the fraction of nominal capacity beyond which a single reading is removed, the tolerance within which a truncated record reaches EOL, the share of units at a parameter bound beyond which a family is not eligible and the margin within which a simpler family is preferred, the factor by which a pattern type must exceed its noise-only rate, the minimum counts per measurement, the longest run of readings removed in a channel other than capacity, the tolerance and the maximum number of iterations of the offset fit, the smoothing window length and polynomial order, the pattern detection thresholds, the number of channels and which of them enter the target, the channel shares β_c and position profiles of the weights w, the definition of the reference point x^0, the largest share of the variance of the decomposable target that the differences in level between units may carry, the window length L, and the generator settings varied in the protocol.

Detecting the inserted patterns and estimating their parameters. Two families of detection procedure are established in battery prognostics, distinguished by what triggers a detection. The first is triggered by the capacity sequence itself: it tests the residual of a fitted degradation trend for a departure the trend does not account for, for instance by a statistical test on the residual sequence [39]. The second is triggered by the rest time between adjacent cycles, on the mechanism that a rest longer than a threshold produces a regeneration over the cycles that follow [35, 36]. The two are not disjoint in the information they use: decomposition-based methods separate the degradation trend from the sequence and then identify regeneration by a rest-time threshold [34]. We adopt the first. Rest time is recoverable from the timestamps of some public datasets and not of others, whereas the residual of the trajectory fit is available for every dataset by step two above, and the same rule has to apply to every profile for the fitted parameters to be comparable across them.

The rule is as follows. A candidate pattern is a run of consecutive positions whose residual from the fitted trajectory exceeds a declared multiple of the residual scale and whose length reaches a declared minimum without exceeding the smoothing window; the sign of the run gives the type. A departure that lasts longer than the smoothing window survives smoothing, so it is part of the smoothed capacity and changes the degradation state, which an inserted pattern must not do; it therefore belongs to the mean component. For each detected pattern we record the position of its extremum, its signed amplitude, and its duration in positions, and we estimate the distribution of amplitude and of duration together with the per-unit rate of occurrence over the dataset. Amplitude and duration are the quantities the rest-time literature also extracts from measured sequences, as the value reached at the regeneration cycles and the number of cycles the regeneration region spans [35], so the fitted distributions are comparable with published characterizations of the same phenomenon [35, 36].

The procedure has two thresholds, and both are declared before fitting and reported with the counts they produce, because the estimated distributions are not insensitive to them: a multiple set too low absorbs noise into the pattern distributions, inflating the rate and biasing the amplitude distribution downward, while one set too high empties them. We therefore report the detected rate and the amplitude distribution over a declared range of the multiple, alongside the residual scale itself, and a type is enabled in a profile only if the rule detects it at least a declared factor more often than in noise simulated with the profile's own fitted variance and autocorrelation, since detections that the fitted noise alone produces are the noise the threshold exists to exclude; a type not detected beyond that rate is not enabled rather than being generated at an assumed rate.

Splitting and use of future information. Windows drawn from one unit are strongly dependent, so all splits are performed at the level of units rather than windows, on both generated and measured data. Normalization statistics are computed on training units alone. No input feature may be derived from a quantity that depends on positions after t or on the full length of a sequence, since that length is the label; in particular the normalized position t/T is excluded while the elapsed position t is not. This restriction applies to model inputs. It does not apply to the fitting procedure above, which is not a model and is allowed to use each measured unit in full, nor to the targets, which are labels.

3.4 Profiles

Three profiles are built and evaluated, and a fourth dataset was fitted as a candidate and is reported but not evaluated. **MATR** [13] supplies the reference channel set, recording discharge capacity, internal resistance, charge time, mean discharge voltage, and temperature statistics per cycle. **HUST** [14] contains cells of the same chemistry and format cycled under different discharge protocols in a different laboratory, and therefore tests whether the fitting procedure transfers when the cell is held fixed and the protocol is not. HUST is the profile against which the fidelity measurements are primarily read: the transfer ratio, the covariance agreement and the property table, but not the discriminative error, which is void for MATR and HUST at their numbers of held-out units. A rule declared before these measurements were made requires such a profile to be named, with the generator left as fitted, if the transfer ratio of MATR lies outside the interval of the HUST ratio; it does, and HUST is named because its interval contains one (Section 4.1). The third profile, **FORM**, is built from the formation dataset of Cui *et al.* [15, 40]: single-crystal NMC532/artificial-graphite pouch cells, of a chemistry, format and laboratory different from those of MATR and HUST, formed under 62 designed formation protocols and a baseline one and then aged under one protocol identical for every cell (a constant-current, constant-voltage charge at 1C to 4.4 V and a constant-current discharge at 3C/4 to 3.0 V, interrupted every 100 aging cycles by a diagnostic block of a pulse test and two low-rate reference tests). Positions are the aging cycles; the diagnostic cycles are never positions, and the aging cycle that follows them is a rest event. Its capacity channel is the discharge capacity of the aging cycle, a full-capacity measurement under one procedure at every position, and its charge time is the duration of the constant-current and constant-voltage charge. FORM replaces NASA PCoE [41], which was included for its documented capacity regeneration but whose held-out split is too small for any fidelity measure to be read, and whose regeneration, at its fitted amplitudes, carries too little of the target for a trained model to use, a limitation FORM's recovery shares (Section 4.2). Like the candidate before it, FORM was selected by declared criteria,

written before any of its files were downloaded but after its documentation, and measured values of one of its cells, had been read (Section 5): a single cell type; a capacity channel that measures full capacity under one procedure at every position; at least 66 units reaching EOL, so that 20 of them are held out, none shorter than $k + L$ positions; capacity and at least two channels that change with degradation in every unit; and a licence that permits releasing fitted parameters. All 182 cells that the dataset's authors document and released reach EOL. Its formation protocols set part of each cell's early capacity and channel levels, a manufacturing variation the profile is meant to carry: on the fitting split, the 61 protocols represented explain 0.41 of the variance of each cell's mean offset term in the decomposable target under recency weighting and 0.41 of that of its early capacity, calibrated against random protocol labels ($p < 0.001$ for both), while the charge-time and mean-discharge-voltage offsets taken one at a time, at 0.24 and 0.16, are not explained at $p < 0.05$; before the profile was fitted, the share of the per-unit offsets in the variance of the decomposable target was measured on the measured units against the bound described below, and it is 0.12. FORM is also the only profile in which the detection rule of Section 3.3 enables an inserted pattern type: its capacity recovers in the aging cycles that follow a diagnostic block, which both rests the cell and discharges it at a lower rate, so the recovery is attributed to the block rather than to rest alone, and it is inserted as a positive pattern at its fitted rate and amplitude. The parameters fitted to FORM and the units generated from it are released under the licence of the dataset, CC BY-NC 4.0. The first candidate for the third profile, **ISU-ILCC** [16], has NMC/graphite lithium-polymer cells that differ in chemistry and format from those of MATR and HUST and were cycled under conditions that differ in charge rate, discharge rate, and depth of discharge. It was selected by the same criteria, with the capacity channel admitted under a declared extraction. Because most of its conditions discharge only part of the cell in each aging cycle, positions are aging cycles whose measured discharge capacity is rescaled by a factor that varies smoothly over the cycle index and is anchored at the periodic full-capacity reference tests, which interrupt the aging sequence and are never positions; the profile is restricted to the cells the dataset's authors validated, in the conditions whose aging cycles discharge at least half of the cell's capacity. The capacity channel therefore combines two procedures, the reference test setting its level and the partial discharge its variation between tests, and a recovery after a reference test may reflect the kinetics of a partial discharge rather than regenerated capacity. The test conditions of ISU-ILCC set the level of its charge time and of its mean discharge voltage, so the per-unit offsets of these channels (Section 3.1) record the condition under which each cell was cycled. A profile is not evaluated when the offsets carry more than half of the variance of the decomposable target, a bound declared before the offsets were fitted, since the target would then record mainly the level at which a unit sits rather than how far it has degraded. ISU-ILCC exceeds that bound; its audit, its fit, and the share of the offsets are reported, and the measurements of Sections 3.5 and 3.6 are made on MATR, HUST and FORM.

Before a profile is built, the property it is included for is verified on the data rather than taken from the literature. We report the fraction of units whose smoothed capacity channel is monotone within tolerance, the detected rate and amplitude of capacity regeneration, the fraction of units reaching the declared EOL threshold, how often the early-life reference differs materially from the first-position capacity, the distribution of unit lengths, and which channels are available across all units. A dataset that does not exhibit the property it was selected for has its role reassigned or is dropped, and the outcome is reported.

3.5 Evaluation Protocol

3.5.1 Profile Fidelity

The first research question asks how closely each profile reproduces the dataset it was fitted to. Two measures established in the generative time-series literature are reported per profile: the classification error of a discriminator trained to separate generated from measured windows [31] and the agreement of the cross-channel covariance structure. Both are computed on windows, with the window length and the effective number of independent units stated alongside. A third, a Fréchet-style distance computed in a learned representation space [32], is not reported, since at the numbers of held-out units available here its interval over units is too wide for it to inform the comparison. Every measurement in the protocol has a declared minimum number of units or events, set so that its resampling interval does not extend beyond half of its value on either side; a measurement below its minimum is reported as void with its count, following the convention of the usability checks. The discriminative error is read by its distance from 0.5, so it is readable only if, for every generation seed, the half-width of its 95% interval over held-out units is at most half of that distance; otherwise its cell is void.

The discriminative measure is reported as a coarse indicator only. On sequences of the lengths considered here, a discriminator separates the two sources whenever any property is mismatched, and its error saturates without indicating which property is responsible. The property-level comparisons therefore carry the evidence: the fitted distribution of θ, the noise variance and autocorrelation, the length distribution, the rate and amplitude of capacity regeneration where a profile carries inserted patterns (the amplitude on the measured side only, since the inserted patterns are drawn from its fitted distribution), and the cross-channel covariance are each reported for the generated and the measured side.

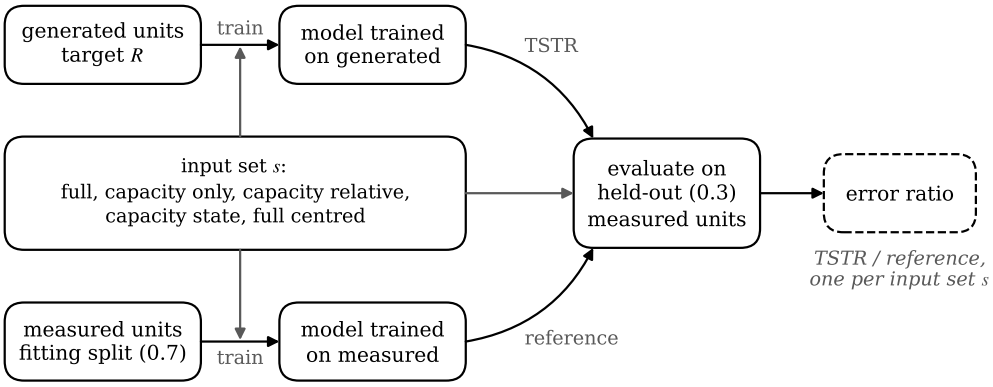


Figure 5: The transfer measurement. Two models are trained, one on a profile and one on the measured units used to fit it, and both are evaluated on measured units held out from fitting. The reported quantity is the ratio of their errors, computed separately for each of five input sets given alike to both models, each with the elapsed position t: all channels (full); capacity alone (capacity only); capacity divided by the unit's early-life reference q_1 (capacity relative); capacity mapped to $(q_1 - q)/(q_1 - \rho\, q_{\mathrm{nom}})$ (capacity state); and all channels, every channel other than capacity centred per unit on its first k positions (full centred).

The transfer measurement follows the TSTR procedure [30] and is shown in Figure 5. A model is trained on a profile using the transfer target of Equation 6, and is evaluated on units of the paired dataset that took no part in fitting the profile, using their RUL label. The same architecture trained on the measured units of the fitting split, and evaluated on the same held-out units, gives the reference. To locate a gap, the ratio is also computed with the input of both models restricted to each of the declared input sets of Figure 5; a difference between the log ratios of two sets is read as holding only if its paired interval over held-out units stays above zero when each generation seed and each initialization seed is left out in turn. We report the ratio of the two errors with a bootstrap interval rather than the absolute error, which is not comparable across datasets with different lifetime ranges. A ratio near one indicates that a model learns as much from the profile as from the dataset it stands for. Each profile is evaluated against its own paired dataset only, since a pooled evaluation would allow a close match on one dataset to compensate for a poor match on another.

3.5.2 Usability of the Attribution Ground Truth

The second research question asks whether the attribution ground truth is usable, which requires that a model can learn the decomposable target, that it uses each channel that enters the target, and, where a profile carries inserted patterns, that the two kinds of contribution can be told apart. Four checks are reported, alongside the two properties of the target declared in Section 3.2. A model must reach a predetermined accuracy on each profile before any attribution is computed on it; scores obtained below that threshold are reported as void rather than as poor. The correlation between the graded and the sparse contribution fields across the dataset must remain below a declared bound, so that a method is not credited for recovering one when it has recovered the other. Retraining with each of the two terms of Equation 4 removed in turn must change model behaviour in the direction the construction predicts, confirming that both are used. Finally, permutation importance must show that models do not rely on the null channels of Section 3.1; since these carry no degradation state by construction, a model that does indicates that the generated data leaks the target through a channel intended to be irrelevant, and the profile is corrected before use. Channels that carry the degradation state but have zero target weight are not held to the same bound, since a model may read the state from them as legitimately as from the channels that enter the target. For each of them we report how well it is predicted from the remaining channels and its importance when it is permuted conditionally on them, which is near zero when the channel is redundant and positive only when it carries information the other channels do not.

The channels that enter the target are held to the reverse requirement: the model must use each of them. Since every channel other than the null channels carries the same degradation state, an accurate model need not read every channel to which $\phi^\star$ assigns part of the target, and where it does not, the channel-level ground truth describes a function the model did not learn. Each weighted channel is therefore permuted conditionally on the remaining channels in the same way as the redundant channels, by replacing it with its ridge-regression prediction from them plus residuals permuted across test windows, and is taken as used if the lower end of the 95% interval over test units of the resulting increase in error lies above zero. The rule is read on the primary trained model of each profile and weighting, on which the methods' scores are reported, and its outcome is listed for every member of the ensemble of Section 3.6. Where a

weighted channel is not used, the scores on trained models that depend on how attribution is divided among channels, namely the share of attribution on zero-weight channels and the channel allocation error defined below, are void for that profile and weighting, with that reason; the temporal profile error is then averaged over the used channels only, and rank agreement, which compares individual cells, is still reported. Whether each weighted channel is used materially, with the lower end of its interval above the tolerance that the leakage test allows a null channel, is reported beside the rule and does not change it.

3.5.3 Responsiveness of the Scores

The third research question asks whether the scores distinguish better from worse attribution maps, and which failures of a map each of them registers. The first part is measured without involving any attribution method. The ground-truth field itself is a correct attribution map by construction, so it can be degraded in controlled amounts: adding noise of increasing variance, shifting a known fraction of the mass toward the start or the end of the sequence, smoothing across neighbouring positions, and randomly permuting a known fraction of the values. A score that does not worsen as the map is degraded is not measuring what it is intended to measure, and the smallest degradation that a score resolves reliably is reported as its resolution, with the mean change of the score at that magnitude. Because a paired interval over many units excludes zero for any consistent change, however small, a score can resolve a change too small to matter. Each score is therefore also read for whether it *registers* a degradation, that is, whether at some magnitude, and at every larger one, it resolves the degradation with a mean change of at least a tenth of the distance between its value on a fully permuted map and its value on the undegraded field; the smallest such magnitude is reported. A tenth of that distance is taken as the smallest change a reader comparing methods would act on. Smoothing is read under recency weighting only, since under uniform weighting the graded field is nearly constant along the window and smoothing leaves it almost unchanged.

Rank agreement, the rank correlation between a map and the graded field over the cells of a window, changes little under two of these degradations, smoothing a map along the window and moving part of each channel's mass to its last position, since on a field that grows steadily toward the end of the window both leave the order of the cells within each channel largely unchanged. Two further scores therefore measure separately how a map A divides attribution among channels and how it distributes attribution along the window. The *channel allocation error* is the total-variation distance $\frac{1}{2}\sum_c |s_c(A) - s_c(\phi^\star)|$ between the shares $s_c(M) = \sum_u |M_{u,c}| / \sum_{u,c'} |M_{u,c'}|$ of absolute attribution on each channel, taken over all channels, those of zero weight included; it is 0 when the shares agree and at most 1. The *temporal profile error* is computed for each weighted channel as the 1-Wasserstein distance over positions between $|A_{\cdot,c}|$ and $|\phi^\star_{\cdot,c}|$, each normalized to unit mass, which for positions one apart is the sum over the window of the absolute difference between their cumulative shares; the distances are averaged over the weighted channels with weights equal to the mass of $|\phi^\star|$ on each. It is measured in positions, is 0 when the profiles agree, and is at most $L - 1$. A window in which a map has no mass, or a weighted channel on which either map has none, is left out of the corresponding average, and such cases are counted. Both scores are computed on the same maps as rank agreement, which is kept, together with the share of attribution on zero-weight channels and, where inserted patterns exist, retrieval of the sparse set.

The second part varies the generator settings that control task difficulty: noise variance, the sharpness of the transition in the degradation state, sequence length, and the amplitude of the inserted patterns. Scores that saturate near their maximum indicate a task too easy to separate methods, and scores near chance indicate one that is too hard. The range over which they remain responsive is reported as the operating range of the benchmark, so that later use does not extend beyond the settings for which the measurements were shown to be informative. This part is not run for the profiles reported here. On an earlier version of the profiles, every setting at which the other measurements are made lay inside the range it gave for rank agreement. Its definition is kept for later use, the earlier results remain available in the released repository, and the two scores introduced above have no measured operating range.

3.6 Models and Baselines

Attribution is evaluated on a single trained architecture, a recurrent network with long short-term memory units, together with the reference model of Section 3.2. The benchmark scores attribution maps against known contributions and is independent of the model used to produce them, so a second trained architecture would not contribute to the three research questions; the reference model provides the comparison that removes model error, which a second trained model provides only indirectly. The two columns differ in kind, and the reference model is the primary scoring target. Its function is known, so its faithful attribution from the reference point is known exactly: $w_{u,c}\,(x_{u,c} - x^0_c) = \phi^\star_{u,c} + w_{u,c}\,\varepsilon_{u,c}$, which differs from $\phi^\star$ only by the weighted noise of the window. Scored against $\phi^\star$, it sets the ceiling of the reference column, and that column measures the attribution method. The channels of a window share one degradation state, so many functions of the window reproduce the decomposable target equally well, and an accurate trained model need not implement the one that defines $\phi^\star$; its column therefore also reflects which of these functions

the model learned. To bound that effect, an ensemble of equally accurate models differing in initialization and size is trained, each of the three methods below is run on every member, TimeSHAP under recency weighting only, and the spread of their scores is reported beside the scores of each method as the limit it places on any method.

Three published methods are run at their default configurations, apart from the baseline, on both the trained and the reference model: TimeSHAP [42] for perturbation-based Shapley attribution over sequences, Integrated Gradients (IG) [43] for gradient-based attribution, and feature occlusion [44]. Methods that optimise a mask per instance [12, 18, 19] are excluded, since their scores depend on hyperparameters whose tuning is a comparison of methods rather than a reference measurement. These results are reported as reference values for later work rather than as an evaluation of the methods themselves. Accordingly, they are reported in aggregate, without stratification by position within the trajectory and without a diagnosis of the causes of any differences observed. All three are run on the same windows of each profile, and TimeSHAP, the only one that samples, with every declared attribution seed.

Each of the three methods computes attributions relative to a baseline, and Shapley-based and path-based attributions sum to the difference between the model output on the window and on that baseline. Scoring them against $\phi^\star$ therefore presumes that the model output on the baseline is zero, which holds for the reference model only at the reference point x^0. The declared baseline is accordingly the pristine window, every position of which is at x^0: it is the Integrated Gradients baseline, the occlusion value, and the TimeSHAP background instance. Unlike a zero input, the pristine window lies close to the first windows of every unit, so the trained model is not queried on a window unlike any it was trained on. Because the decomposable target is affine with known weights, the attribution ground truth from any other baseline b is $w_{u,c}\,(m_{u,c} + \sum_k p_{k,u,c} - b_{u,c})$, and the benchmark supplies it, so a method run from a different baseline is scored against the ground truth that matches it and methods can still be run at their defaults. TimeSHAP is additionally run from the average event it uses by convention, as a readout of whether the ordering of the reference values depends on the baseline. A background conditioned on the local mean component is not adopted as the reference, although it is natural on a drifting channel, since it would make trend-conditional backgrounds correct by construction and any later evaluation of such backgrounds circular. Occlusion satisfies no efficiency property, so its values are compared with $\phi^\star$ in rank and retrieval but not in sum.

Generation, model initialization, and any sampling performed by an attribution method are controlled by separate random seeds, so that the contribution of each to the variance of a score can be reported separately. The generator, the fitted profile parameters, the seeds, the trained and reference models, the ground-truth fields, and the scoring code are made available as stated in Section 1.

4 Results

This section reports the measurements defined in Section 3.5, one subsection per research question, followed by the reference values of the published attribution methods of Section 3.6. Intervals are obtained by bootstrap over units, and the seeds controlling generation, model initialization, and attribution sampling are varied separately so that their contributions to the spread are reported apart. A measurement whose count of units or events falls below its declared minimum, or whose interval is too wide to be read, is reported as void, with the count and the reason in the table note.

Three profiles are evaluated. MATR rests on the 135 cells of the three batches of [13], of which 128 reach EOL, HUST on all 77 cells of [14], and FORM on all 182 aged cells of [15]; every cell of HUST and FORM reaches EOL. The fitting splits hold 95, 54 and 127 cells and the held-out splits 40, 23 and 55. ISU-ILCC is not evaluated: all 145 of its cells in scope reach EOL, and its fit selects the power law on the 102 cells of its fitting split, but its per-unit offsets carry 0.765 to 0.823 of the variance of the decomposable target over the three generation seeds, above the declared bound of 0.5; its test conditions account for 99% of the variance of its charge-time offsets and 98% of that of its mean-discharge-voltage offsets (Section 3.4). In FORM the same share is 0.12 on the measured cells, read before the profile was fitted, and 0.11 to 0.13 over all windows of its generated units at the three generation seeds (0.159 on the test windows of Table 4).

The profiles differ in whether they carry inserted patterns. On the fitting splits of MATR and HUST, the positive capacity-residual runs occur at 0.85 and 0.47 times the rate at which the same rule detects runs in noise with the fitted variance and autocorrelation, and the negative runs at 0.78 and 0.98 times that rate, below the declared factor of 2 that enables a type, so neither profile carries inserted patterns and the graded field is its whole attribution ground truth. In FORM the positive runs occur at 2.28 times that rate, 312 runs over 127 cells with a median amplitude of 2.7 mAh and a median duration of 4 positions, and the type is enabled; the negative runs, at 0.96 times, are not. Positive runs start at a rest event or within the ten positions after it 4.57 [3.99, 5.18] times as often as at random positions of the same series, and within the eleven positions before one 0.98 [0.63, 1.34] times. Of these 1 114 rest events, 1 073 follow a diagnostic block; after the other 41 the ratio is 2.04 [0, 4.51], so the recovery is attributed to the diagnostic block, which both rests the cell and discharges it at a lower rate, rather than to rest alone (Section 3.4). Recency weighting is primary and uniform weighting a sensitivity readout; final-position weighting is not scored. Where a comparison shows what the

per-unit offsets and the correlated noise change, the profile without them, fitted to the same splits, is reported beside the profile with them (Tables 1 and 3).

4.1 Profile Fidelity

Table 1 reports, per profile, the discriminative measure, the covariance agreement and the transfer ratio of the profile, of the same profile drawn with independent noise, and, for MATR and HUST, of the profile without per-unit offsets and correlated noise. Table 3 locates the transfer gap by restricting the input of both models, and Table 2 compares the generated with the measured side of each fitted quantity. The discriminative measure is a coarse indicator only (Section 3.5), and Figure 6 is an illustration.

On the profiles without per-unit offsets and correlated noise, the transfer gap of MATR sat in the channels other than capacity. A model trained on that profile reached 3.42 [2.28, 4.96] times the error of the same architecture trained on measured cells, against 1.46 [1.11, 1.76] for HUST. Given capacity and elapsed position alone, MATR transferred at 1.10 [0.86, 1.50], and the difference of log ratios, 1.14 [0.66, 1.75], held when each seed was left out (Table 3). With the other channels added, the model trained on measured cells lowered its error from 326 to 151 cycles, and the model trained on generated units raised it from 357 to 516. Centring each of those channels per unit removed 0.52 [0.21, 0.92] of the 1.14 log units, a difference that also held, so constant per-unit levels explained about half of the gap. Dividing capacity by the unit's early-life reference changed nothing, -0.01 [-0.32, 0.29], so the spread of initial capacity between cells was not shown to contribute. None of these differences held for HUST, whose cells share one charging protocol.

With per-unit offsets and correlated noise, the MATR profile transfers at 1.49 [1.04, 2.29], down from 3.42, and the HUST profile at 1.17 [0.94, 1.34], down from 1.46 (Table 1). Over the fifteen seed combinations the ratio ranges from 1.23 to 2.14 for MATR and from 1.03 to 1.38 for HUST. HUST's interval contains one: a model trained on the profile is not distinguished from one trained on the measured cells. MATR's gap is not closed: 1.49 lies outside HUST's interval [0.94, 1.34]. The gain comes from the offsets. Drawn with independent noise, at one generation seed, the profiles transfer at 1.60 [1.06, 2.50] and 1.12 [0.92, 1.29], against 1.58 and 1.15 with correlated noise at that seed; the offsets lower the ratio by a factor of 2.19 [1.69, 2.79] in MATR, against 3.51 for the profile without offsets with the measured cells cleaned as for the others (Table 3), and 1.30 [1.03, 1.60] in HUST, and the correlated noise by 1.02 [0.92, 1.10] and 0.98 [0.88, 1.06], that is, not at all. For MATR, capacity alone now transfers at 0.98 [0.92, 1.08], but the difference of log ratios, 0.42 [0.06, 0.90], no longer holds when a seed is left out, and centring the other channels per unit no longer lowers the ratio (1.93 [1.41, 2.58]). The remaining gap of MATR is therefore not located robustly; the point estimates place it in the other channels, in structure that is neither a constant level nor, since the profile reproduces it, the correlation of their noise. For HUST no difference holds: the one interval above zero, all channels against offsets removed, 0.19 [0.01, 0.36], does not stay above zero when a seed is left out.

FORM transfers at 1.54 [1.28, 1.74], from 1.42 to 1.78 over the fifteen seed combinations. Like MATR's, its ratio lies outside HUST's interval, although the two intervals overlap, and its own interval excludes one. Unlike MATR's, its gap is located, and in the same place as MATR's gap on the profile without offsets. Given capacity and elapsed position alone, FORM transfers at 1.08 [0.99, 1.15]. With the other channels added, the model trained on measured cells lowers its error from 135 to 94 cycles, while the model trained on generated units stays at 147 and 145, and the difference of log ratios, 0.35 [0.17, 0.48], holds when each seed is left out. Centring the other channels per unit on their first k positions removes 0.28 [0.10, 0.40] of it, a difference that also holds, and dividing capacity by the early-life reference again changes nothing, -0.03 [-0.10, 0.01]. The gap of FORM is therefore in the channels other than capacity, and most of it in their levels early in a cell's life: those levels tell a model about the remaining life of a measured cell, and much less about that of a generated one.

Table 2 shows that the trajectory is reproduced: generated against fitting and held-out cells, the median drift is 23.9 against 25.6 and 25.6 mAh per 100 cycles for MATR, 17.0 against 16.8 and 16.3 for HUST, and 7.87 against 7.78 and 8.27 for FORM; the transition falls at the same fraction of life; and the median unit length is 823 against 770 and 744 cycles, 1807 against 1860 and 1938, and 683 against 701 and 665. The family fits generated units, which are drawn from it, with less error than measured ones. The offsets of the other channels are reproduced: the interquartile range of the charge-time offsets is 4.96 against 4.77 min for MATR, whose held-out cells, at 1.30, span fewer charging-policy groups, 1.97 against 1.68 for HUST, and 1.31 against 1.43 for FORM, and that of the mean-discharge-voltage offsets of FORM 12.9 against 13.3 mV. So is the within-unit residual correlation, which departs from the fitting split by 0.11, 0.03 and 0.13, against 0.09, 0.03 and 0.10 for the held-out cells; the excess of MATR and FORM lies in the capacity pairs, whose estimate is biased. FORM's positive residual runs, inserted patterns and noise together, occur at 0.27 per 100 positions against 0.33 and 0.29. Of the variances and spreads, five depart from the fitting split by a quarter or more: the capacity noise variance of MATR, 1.88 against 2.82 and 4.57 mAh^2, and of FORM, 0.99 against 1.43 and 1.56 mAh^2; the charge-time noise variance of HUST, 1.02 against 1.39 and 1.43 min^2; and the spread of the early-life reference of

Table 1: Profile fidelity, on windows of $L = 24$ positions. One block per profile, one row per generator: the generator of Section 3.1, with per-unit offsets and correlated noise; the same profile drawn with independent noise per channel, at generation seed 0; and, as an ablation for MATR and HUST, the generator without per-unit offsets and with independent noise, fitted to the same splits. Units: held-out measured units and generated units per generation seed; the transfer ratio is evaluated on the held-out units that reach EOL, 38 of the 40 in MATR, all 23 in HUST and all 55 in FORM. Discriminative error: range over three generation seeds, 0.5 meaning indistinguishable; in the two lower rows of a block it was not assessed for readability and is read only against the per-seed values in the notes, not as a level. Covariance agreement: relative Frobenius distance between the channel correlation matrices of generated and held-out windows, mean over generation seeds, with the same distance between the measured fitting and held-out windows, and in the first row of a block its 95% unit-bootstrap interval, in parentheses. Transfer ratio: the error of a model trained on the profile divided by that of the same architecture trained on the measured fitting split, both evaluated on the held-out measured units, pooled over the generation seeds of the row and five initialization seeds, with its 95% BCa interval; a ratio near one indicates that a model learns as much from the profile as from the dataset it stands for. Shares in transfer, at generation seed 0, as ratios of transfer ratios with paired 95% BCa intervals over held-out units: of the per-unit offsets, the profile without offsets (with the measured cells cleaned as for the others; Table 3) over the profile with offsets and independent noise; of the correlated noise, the profile with independent noise over the profile with both. MATR: offsets 2.19× [1.69, 2.79], correlated noise 1.02× [0.92, 1.10], with the profile with both at 1.58 [1.13, 2.37] at that seed; HUST: offsets 1.30× [1.03, 1.60], correlated noise 0.98× [0.88, 1.06], with the profile with both at 1.15 [0.91, 1.31] at that seed. [a]MATR: 95% unit-bootstrap half-width above half the distance from 0.5 for 2 of 3 generation seeds; values 0.33 [0.19, 0.48], 0.24 [0.15, 0.36], 0.42 [0.28, 0.56]. [b]HUST: 95% unit-bootstrap half-width above half the distance from 0.5 for 2 of 3 generation seeds; values 0.37 [0.27, 0.48], 0.46 [0.38, 0.54], 0.62 [0.58, 0.66]. [c]FORM: the transfer ratio with independent noise is computed for MATR and HUST only, on their existing sets.

Profile	Generator	Units	Discriminative err.	Covariance agr.	Transfer ratio [CI]
MATR	offsets, correlated noise	40 / 300	void[a]	0.21 (0.17; 0.08–0.44)	1.49 [1.04, 2.29]
	independent noise		0.36	0.21	1.60 [1.06, 2.50]
	no offsets, independent noise		0.05–0.09	0.69 (0.28)	3.42 [2.28, 4.96]
HUST	offsets, correlated noise	23 / 300	void[b]	0.05 (0.04; 0.02–0.12)	1.17 [0.94, 1.34]
	independent noise		0.41	0.06	1.12 [0.92, 1.29]
	no offsets, independent noise		0.12–0.20	0.42 (0.04)	1.46 [1.11, 1.76]
FORM	offsets, correlated noise	55 / 300	0.10–0.17	0.35 (0.30; 0.20–0.43)	1.54 [1.28, 1.74]
	independent noise		0.09	0.32	void[c]

HUST, 26.9 against 21.2 and 19.6 mAh, and of FORM, 3.43 against 2.63 and 2.31 mAh. The charge-time noise variance of FORM departs by just under a quarter, 0.58 against 0.77 and 0.62 min^2, and lies close to that of the held-out cells.

The cross-channel correlation of generated windows departs from that of held-out windows by 0.21 in relative Frobenius distance for MATR, 0.05 for HUST and 0.35 for FORM; each lies within the interval of that distance between measured fitting and held-out windows, [0.08, 0.44], [0.02, 0.12] and [0.20, 0.43], so the covariance gap is closed on all three profiles, against 0.69 and 0.42 for MATR and HUST without offsets. Drawn with independent noise, the distances are 0.21, 0.06 and 0.32, so the correlated noise contributes nothing measurable here either. The discriminator cells of MATR and HUST are void: the discriminative error is readable for one generation seed only. Relative to the profiles without offsets, its mean over seeds rose from 0.07 to 0.33 for MATR and from 0.16 to 0.49 for HUST, by more than the half-ranges over seeds of the profiles without offsets, 0.02 and 0.04, the declared reading of a narrowing gap; neither mean is a readable level. FORM's cell is the only readable one: a classifier separates its generated from its measured windows at an error of 0.10 to 0.17 over the three generation seeds, every interval within the readability bound, and at 0.09 with independent noise. It does not say which property it reads; the transfer gap and the departures of Table 2 are candidates. HUST, whose transfer interval contains one and whose covariance gap is closed, therefore anchors the fidelity of the benchmark; MATR and FORM are reported with their remaining transfer gaps.

4.2 Usability of the Attribution Ground Truth

Table 4 reports the checks of Section 3.5 under recency weighting, together with the share of the variance of the decomposable target carried by the per-unit offsets and the association of the target with RUL. A profile that fails the accuracy threshold has its attribution scores reported as void rather than as poor, and the same convention is used in every table that follows.

The decomposable target is learnable on all three profiles. The primary model predicts it on generated test units with a normalized error of 0.091 for MATR, 0.095 for HUST and 0.179 for FORM, against the bound of 0.30, and all ten ensemble members pass, under uniform weighting as well. The errors of MATR and HUST are higher than the 0.036

Table 2: Property-level comparison between each profile and the dataset it was fitted to: the measured value on the fitting split and on the held-out split, and the generated value of the same quantity, each estimated by the same fitting procedure; medians where a distribution is summarized. Residual correlation: relative Frobenius distance of the within-unit residual correlation matrix from that of the fitting split, and the mean absolute difference of its entries over the pairs that include capacity and over the other pairs. Generated series are read with the record-end tolerance applied to measured records, so a generated unit enters the rows that depend on EOL when it reaches the threshold within that tolerance; in FORM, 21 of the 300 generated units end more than that tolerance, 2.40 mAh, above the threshold, since capacity noise varies slowly, and are left out of those rows. Those rows rest on 90, 38 and 300 in MATR; 54, 23 and 300 in HUST; 127, 55 and 279 in FORM units (fitting, held-out, generated). The capacity-pair correlations carry an errors-in-variables bias of the estimator (the degradation state is read from noisy capacity), once in the fitted target and again when re-estimated on generated data. Residual runs are counted by the detection rule of Section 3.3; only FORM carries an inserted pattern type, so in MATR and HUST every run counted on generated units comes from noise.

	MATR			HUST			FORM		
Property	fitting	held-out	gen.	fitting	held-out	gen.	fitting	held-out	gen.
Fit error of the profile's family [fraction of q_1]	0.0061	0.0054	0.0033	0.0027	0.0032	0.0014	0.0105	0.0109	0.0077
Drift rate [mAh/100 cyc.]	25.6	25.6	23.9	16.8	16.3	17.0	7.78	8.27	7.87
Transition position [$\times T$]	0.999	0.999	0.999	0.686	0.651	0.676	0.999	0.998	0.999
Unit length T [cycles]	770	744	823	1860	1938	1807	701	665	683
Early-life reference q_1, IQR [mAh]	10.8	10.4	12.4	21.2	19.6	26.9	2.63	2.31	3.43
Noise variance, capacity [mAh^2]	2.82	4.57	1.88	1.10	1.06	0.945	1.43	1.56	0.989
Noise variance, charge time [min^2]	0.694	0.799	0.584	1.39	1.43	1.02	0.765	0.624	0.575
Noise autocorrelation, charge time	0.89	0.81	0.88	0.99	0.99	0.99	0.95	0.95	0.94
Offset IQR, charge time [min]	4.77	1.30	4.96	1.68	1.62	1.97	1.43	1.21	1.31
Offset IQR, mean discharge V [mV]	59.0	53.2	63.8	53.0	47.4	54.2	13.3	8.48	12.9
Positive residual runs per 100 positions	0.12	0.12	0.09	0.06	0.04	0.11	0.33	0.29	0.27
Residual correlation, rel. Frob. to fitting split	0.00	0.09	0.11	0.00	0.03	0.03	0.00	0.10	0.13
capacity pairs, mean $\lvert\Delta r\rvert$	0.000	0.026	0.075	0.000	0.023	0.029	0.000	0.065	0.109
other pairs, mean $\lvert\Delta r\rvert$	0.000	0.059	0.056	0.000	0.012	0.013	0.000	0.085	0.094

Table 3: Transfer ratio by model input, locating the transfer gap, on the profiles without per-unit offsets and correlated noise (MATR and HUST) and on the profiles of Table 1. Each input set, which always includes the elapsed position t, is given to both the model trained on generated units and the model trained on measured units; ratios with 95% BCa intervals over held-out units. Full: all channels; capacity relative: capacity divided by the unit's early-life reference q_1; capacity state: capacity mapped to $(q_1 - q)/(q_1 - \rho\, q_{\mathrm{nom}})$; full centred: all channels, every channel other than capacity centred per unit on its first k positions. Lower rows: differences of log ratios between input sets, with 95% paired intervals over held-out units; a difference is read as holding only if its interval stays above zero when each generation seed and each initialization seed is left out in turn. The columns without offsets use the measured cells as cleaned before the removal of readings that no degradation produces was extended, in channels other than capacity, to runs of up to three readings; with that extension, as in the columns with offsets, the ratio on all channels without offsets is 3.51 [2.30, 4.99] for MATR and 1.46 [1.11, 1.76] for HUST.

	Without offsets		With offsets		
Input set	MATR	HUST	MATR	HUST	FORM
Full	3.42 [2.28, 4.96]	1.46 [1.11, 1.76]	1.49 [1.04, 2.29]	1.17 [0.94, 1.34]	1.54 [1.28, 1.74]
Capacity only	1.10 [0.86, 1.50]	1.28 [1.08, 1.53]	0.98 [0.92, 1.08]	1.04 [0.98, 1.13]	1.08 [0.99, 1.15]
Capacity relative	1.11 [0.97, 1.24]	1.16 [0.92, 1.45]	1.08 [0.95, 1.21]	1.11 [0.92, 1.35]	1.12 [1.08, 1.17]
Capacity state	1.11 [0.95, 1.30]	1.14 [0.90, 1.40]	1.08 [0.93, 1.25]	1.12 [0.88, 1.37]	1.10 [1.06, 1.15]
Full centred	2.04 [1.41, 2.76]	1.22 [1.05, 1.45]	1.93 [1.41, 2.58]	0.97 [0.85, 1.17]	1.16 [1.05, 1.23]
Δ log: full vs capacity only	1.14 [0.66, 1.75]	0.13 [−0.09, 0.34]	0.42 [0.06, 0.90]	0.12 [−0.07, 0.26]	0.35 [0.17, 0.48]
Δ log: full vs full centred	0.52 [0.21, 0.92]	0.18 [−0.07, 0.40]	−0.26 [−0.59, 0.11]	0.19 [0.01, 0.36]	0.28 [0.10, 0.40]
Δ log: capacity only vs capacity relative	−0.01 [−0.32, 0.29]	0.10 [−0.14, 0.34]	−0.10 [−0.23, 0.03]	−0.07 [−0.28, 0.11]	−0.03 [−0.10, 0.01]

and 0.012 reached without offsets, since the target now includes per-unit levels that a model can read only from the noisy channels that carry them. The offsets carry 0.128 of the variance of y in MATR, 0.343 in HUST and 0.159 in FORM, below the declared bound of 0.5, so all three profiles are taken through the protocol; in HUST, about a third of the variance of y thus records the level at which a cell sits rather than how far it has degraded. The association with RUL weakens accordingly, to rank correlations of -0.71 and -0.61 in MATR and HUST from -0.74 and -0.91 without offsets, and it is -0.69 in FORM. The leakage test passes: permuting either null channel raises the mean squared error, normalized by the variance of y, by at most 0.0003 at the upper end of its interval in any profile under recency weighting, and at most 0.0008 under uniform weighting, against the tolerance of 0.02.

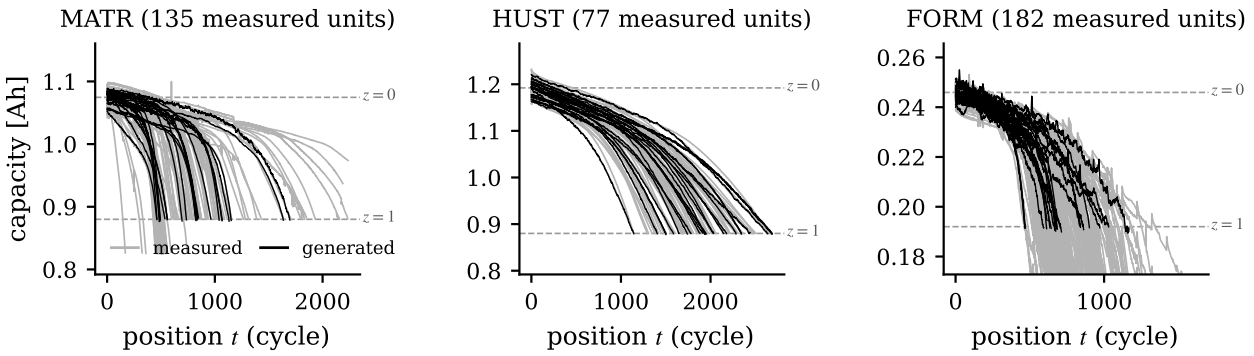


Figure 6: Fitted against measured trajectories, reported as illustration alongside the measurements of Tables 1 and 2: capacity of every measured unit of the profile's scope (grey) and of 25 generated units (black), with the pristine level $z = 0$ and the EOL threshold $z = 1$ of Equation 3 marked. Measured records are drawn to their end, which in FORM lies well past EOL; generated units end at EOL.

The per-unit offsets change which channels a trained model uses. On the profiles without offsets the models read capacity alone: permuting charge time or mean discharge voltage, which carry two thirds of the target weights, raised the normalized squared error by less than 0.001. With offsets, permuting charge time raises it by 0.256 in MATR, permuting mean discharge voltage raises it by 0.808 in HUST, more than capacity at 0.525, and in FORM the two raise it by 0.151 and 0.243 against 0.876 for capacity. An offset changes y through its channel's weight and does not follow from the degradation state, so capacity cannot supply it. Conditionally on the other channels, every weighted channel is used by the primary model of every profile and by all ten members, under both weightings, so no channel-level score is void. Use is uneven, however. In MATR and FORM every weighted channel is used materially, with lower bounds of 0.037 and 0.074 and above, whereas in HUST charge time is used at 0.015 [0.012, 0.018], below the materiality reading of 0.02, and mean discharge voltage at 0.630 [0.545, 0.744], so in HUST part of the ground truth lies on a channel the model reads only weakly. That a channel is used does not fix how a model divides the target among channels, which Section 4.4 reports against the identifiability floor. The redundant channels of MATR add little: internal resistance and temperature are predicted from the other channels with coefficients of determination of 0.75 and 0.47, and their conditional importance is at most 0.0043. HUST and FORM have no redundant channel.

The checks that concern the sparse set are void in MATR and HUST, which carry no inserted patterns, and are read in FORM. There the sparse set is separable from the graded field, with a correlation of 0.045 across test windows under recency weighting and 0.008 under uniform weighting, against the bound of 0.3, but a trained model is not shown to learn it. At its fitted rate and amplitude the pattern term carries 1.7×10^{-5} of the variance of the mean term, far below the declared band of 0.05 to 0.5, and removing it from the training target leaves the model's error on y unchanged, 0.0000 [-0.0001, 0.0002] under recency weighting and 0.0003 [-0.0006, 0.0012] under uniform weighting, whereas removing the mean term raises it by 2.58 [2.23, 3.03]. Under the declared rule, the scores of the sparse set on trained models of FORM are therefore void. The amplitude is not raised to make them readable, since a profile whose patterns exceed those of its dataset no longer stands for that dataset. The sparse set is scored where the map is known to contain it: on the reference model (Section 4.4) and in the degraded-map test of Section 4.3.

4.3 Responsiveness of the Scores

The third research question is answered without any attribution method, by degrading the attribution ground-truth field itself in controlled amounts on 900 windows, 20 from each of the 45 test units of each profile, under recency and uniform weighting. For recency weighting, Figure 7 plots rank agreement and the temporal profile error against the magnitude of each degradation, and Table 5 reports, per score and degradation, the smallest magnitude at which the score registers it; the magnitudes under uniform weighting are given below. Retrieval of the sparse set is measured in FORM, the one profile that carries inserted patterns (Table 4).

Every score takes its expected value on the undegraded field and on a fully permuted map. The undegraded field scores exactly 1 in rank agreement and 0 in both errors, in every profile and under both weightings. Under recency weighting a fully permuted map, which sets the chance level, scores 0.002 in rank agreement in MATR and HUST and -0.001 in FORM, 0.632, 0.464 and 0.476 in channel allocation error, and 5.43, 5.05 and 5.21 positions in temporal profile error. To register a degradation, a score must therefore change by about 0.10 in rank agreement, 0.063, 0.046 and 0.048 in allocation error, and 0.54, 0.50 and 0.52 positions in temporal profile error.

Table 4: Usability of the attribution ground truth, per profile, under recency weighting on generated test units. Model error: normalized root-mean-square error of the primary model, against the declared bound of 0.30, and the number of the ten ensemble members that pass it. Offset term: the share of the variance of y carried by the per-unit offsets, against the declared bound of 0.5. Permutation importance and channel use: increase in mean squared error, normalized by the variance of y, when a channel is permuted across windows or resampled conditionally on the other channels; channel use with 95% intervals over test units, a channel counting as used when the lower bound is above zero. Leakage: upper bounds of the permutation importance of the two null channels, against the tolerance of 0.02. Redundant channels carry the degradation state but have zero target weight; for each, the coefficient of determination when it is predicted from the other channels, and its conditional importance. Pattern term: the variance of the pattern term of Equation 4 relative to that of the mean term, against the declared band of 0.05 to 0.5, and the increase in error on y when it is removed from the training target, with its 95% interval over test units; graded/sparse correlation: across test windows, against the declared bound of 0.3. Generated test units: 45 (MATR), 45 (HUST), 45 (FORM). [a]no inserted pattern type is enabled in the profile.

Quantity	MATR	HUST	FORM
Accuracy, NRMSE of the primary model (members passing)	0.091 (10/10)	0.095 (10/10)	0.179 (10/10)
Offset term, share of Var(y)	0.128	0.343	0.159
Spearman correlation of y with RUL	-0.71	-0.61	-0.69
Permutation importance, capacity	0.908	0.525	0.876
Permutation importance, charge time	0.256	0.049	0.151
Permutation importance, mean discharge V	0.085	0.808	0.243
Channel use (conditional importance), capacity	0.076 [0.066, 0.093]	0.068 [0.057, 0.084]	0.146 [0.128, 0.181]
Channel use (conditional importance), charge time	0.053 [0.037, 0.092]	0.015 [0.012, 0.018]	0.094 [0.074, 0.120]
Channel use (conditional importance), mean discharge V	0.049 [0.038, 0.079]	0.630 [0.545, 0.744]	0.096 [0.076, 0.122]
Members using every weighted channel	10/10	10/10	10/10
Leakage, null channels (upper bound, flat / permuted)	0.0001 / 0.0000	0.0000 / 0.0000	0.0003 / 0.0003
Redundant channels: R^2, conditional importance	IR 0.75, 0.0026; temp. 0.47, 0.0043	none	none
Pattern term, variance ratio to the mean term	void[a]	void[a]	1.7×10^{-5}
Pattern term removed, NMSE increase	void[a]	void[a]	0.0000 [-0.0001, 0.0002]
Graded/sparse correlation $\lvert r \rvert$	void[a]	void[a]	0.045

Resolution alone does not separate the scores. Every score resolves every degradation at the smallest magnitude on its grid, in every profile and under both weightings, including mean changes as small as 2×10^{-7} in the allocation error under smoothing: a paired interval over 45 units excludes zero for any consistent change, however small. Resolution therefore says only that a score moves, and the registering magnitudes show which failures of a map it detects.

The temporal profile error registers every degradation in every profile (Table 5). It registers mass moved toward the start at a fraction of 0.05 and toward the end at 0.1, where it has risen by 0.65 positions in all three profiles (0.32, 0.33 and 0.32 at 0.05), and smoothing at a standard deviation of eight positions, where it reaches 1.38 positions in MATR and FORM and 1.35 in HUST. Rank agreement registers neither positional failure anticipated in Section 3.5. It registers no smoothing on the grid, changing by at most 0.004 at eight positions, and moving 80% of each channel's mass to the end leaves it at 0.999; only at a fraction of 1 does it register a shift, falling to 0.23, 0.26 and 0.26 toward the end and to 0.09, 0.03 and 0.03 toward the start in MATR, HUST and FORM. It does register added noise, at 0.05 in MATR and 0.2 in HUST and FORM, and permuted cells at a fraction of 0.2. The channel allocation error registers the two degradations that move mass between channels, added noise and permutation, at 0.1, and by construction nothing positional: no shift or smoothing changes it by more than 0.003. The two errors thus separate what rank agreement conflates. A map smoothed by eight positions, or with a tenth or more of its mass, short of the whole, moved to the end of the window, passes rank agreement but not the temporal profile error, and the allocation error isolates how attribution is divided among channels from how it is distributed along the window.

Under uniform weighting rank agreement and the allocation error register the same degradations, at the same or the neighbouring magnitude on the grid. A fully permuted map then has a temporal profile error of 2.59 positions in MATR, 1.70 in HUST and 1.77 in FORM, and this error registers mass moved toward either end at the smallest fraction, 0.05. Smoothing registers on no score, as Section 3.5 anticipates: the uniformly weighted field is nearly constant along the window, and smoothing at eight positions moves the temporal profile error by 0.05 positions in MATR, 0.02 in HUST and 0.06 in FORM, against the 0.26, 0.17 and 0.18 needed to register. This is why smoothing is read under recency weighting only.

Retrieval of the sparse set behaves as its pairing intends. It is measured in FORM on 640 windows that contain at least one cell of the sparse set, 20 from each of the 32 test units that have one. Scored on the window paired with its pattern-free counterpart, the undegraded field retrieves the sparse set perfectly, with an average precision of 1 against a chance level of 0.07. Added noise and permuted cells lower it from the smallest magnitude on, to 0.37 at a noise of 0.05 standard deviations of the map and to 0.29 when 5% of the cells are permuted, since the patterns are small against the

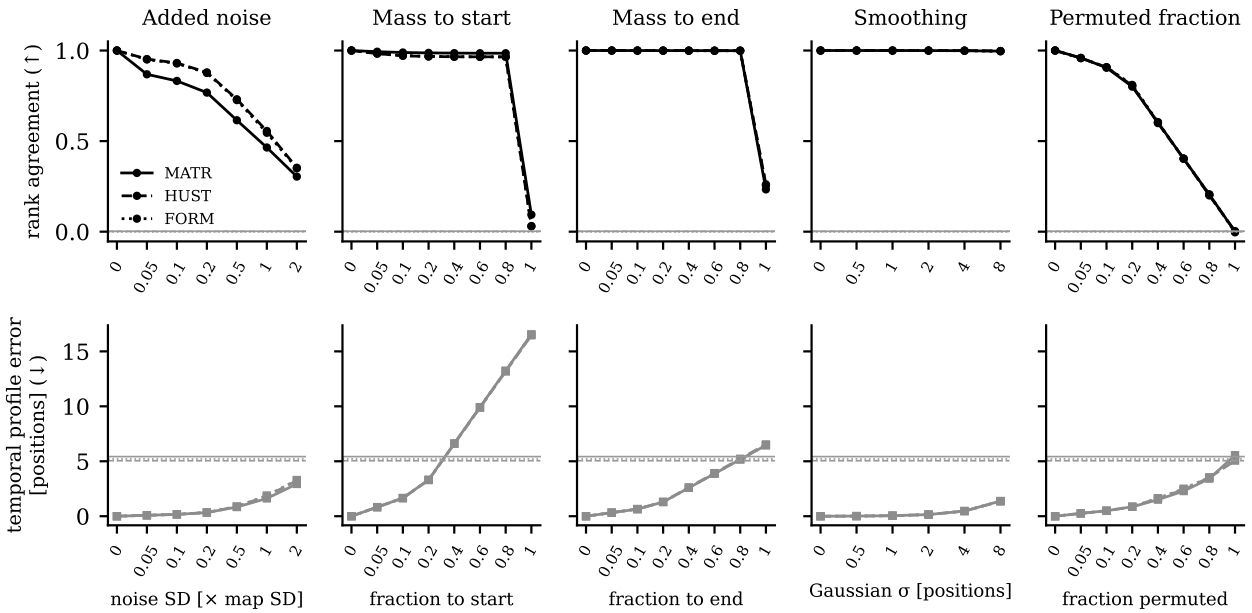


Figure 7: Score response to controlled degradation of a correct attribution map, under recency weighting, averaged over test units: rank agreement against the graded field (top, higher is better) and the temporal profile error in positions (bottom, lower is better), MATR solid, HUST dashed and FORM dotted. Thin grey horizontal lines, in the line style of each profile, mark the score of a fully permuted map; the registering magnitudes are those of Table 5.

Table 5: Registering magnitude of each score under recency weighting: the smallest degradation magnitude at which the score resolves the degradation and its mean change toward chance reaches 10% of the distance between chance and a perfect map, at that and every larger magnitude on the grid; – where no magnitude on the grid registers. A magnitude is resolved when the 95% bootstrap interval over units of the mean change excludes zero in the direction of degradation at that and every larger magnitude. Units of magnitude: added noise in multiples of the standard deviation of the map; shifted mass as the fraction moved; smoothing as the Gaussian standard deviation in positions; permuted fraction of cells. Rank: rank agreement; alloc.: channel allocation error; temporal: temporal profile error. Computed on 900 windows from the 45 generated test units of each profile; in MATR, 2 of the 2 700 channel-windows are left out of the temporal profile error at every magnitude, because the graded field has no mass on them, and none in HUST or FORM.

	MATR			HUST			FORM		
Operator	rank	alloc.	temporal	rank	alloc.	temporal	rank	alloc.	temporal
Added noise	0.05	0.1	0.5	0.2	0.1	0.5	0.2	0.1	0.5
Mass to start	1.0	–	0.05	1.0	–	0.05	1.0	–	0.05
Mass to end	1.0	–	0.1	1.0	–	0.1	1.0	–	0.1
Smoothing	–	–	8	–	–	8	–	–	8
Permuted fraction	0.2	0.1	0.2	0.2	0.1	0.2	0.2	0.1	0.2

graded field, to which both degradations are scaled. Moving mass lowers it gradually, to 0.74 and 0.75 when 40% of each channel's mass is moved toward the end and the start and to 0.10 and 0.11 when all of it is, and smoothing barely moves it, to 0.993 at eight positions, since the difference between two smoothed maps still peaks where the patterns are. Scored on the window alone, the undegraded field reaches 0.095 against a chance level of 0.07, the near-chance value that motivates the pairing (Section 3.2). Under uniform weighting the paired score responds in the same way.

4.4 Reference Values for Published Methods

Table 6 reports the three methods of Section 3.6, run from the declared baseline on the trained model and on the reference model of Section 3.2 under recency weighting, and Figure 8 sets the trained-model scores against the identifiability floor of the ten equally accurate models. As Section 3.6 states, they are reference values for later work, reported without a diagnosis of causes. The reference column is free of model error; the trained column also reflects which of many

equally accurate functions the model learned. In FORM, which carries inserted patterns, every method also explains the pattern-free counterpart of each window, so that the sparse set can be scored on the pair.

On the reference model all three methods return the exact attribution $w(x - x^0)$, with a largest absolute difference over the three profiles of 7.5×10^{-8} for IG, 1.8×10^{-7} for feature occlusion and 2.5×10^{-5} for TimeSHAP, against largest map values of 0.02 to 0.08. This checks the implementation, not the methods: for the affine reference model each of them equals $w(x - x^0)$ by construction, TimeSHAP up to its sampling error. The reference column is therefore the ceiling of the measurement: 0.869 in MATR, 0.957 in HUST and 0.849 in FORM in rank agreement, 0.075, 0.063 and 0.125 in channel allocation error, and 0.38, 0.18 and 0.63 positions in temporal profile error. It is below a perfect score because the exact attribution also carries the window's weighted noise, which $\phi^\star$ excludes. In FORM the difference between each method's maps on a window and on its counterpart retrieves the sparse set exactly, with an average precision of 1 on the 11 test units whose sampled windows contain it: on the reference model, where the pattern term is part of the function, every method localizes it.

On the trained model the methods lie below the ceiling, at 0.420–0.493 in rank agreement in MATR, 0.734–0.750 in HUST and 0.744–0.771 in FORM. The floor bounds how their differences may be read: an ordering of the methods is read only where the largest difference between two of them exceeds the spread of every method over the ten models. In rank agreement that difference, 0.073 in MATR, 0.016 in HUST and 0.027 in FORM, is smaller than the spread of each method, 0.10–0.12, 0.15–0.21 and 0.32–0.34, so a change of trained model moves a method's score more than the choice of method does. In FORM the ten models disagree most: each method spans about a third of the scale in rank agreement, from 0.44 to 0.78 for IG, and the primary model lies near the top of that range. The temporal profile error in all three profiles and the allocation error in HUST and FORM fail the test as well; in FORM the largest allocation difference, 0.050 between feature occlusion and IG, falls short of the spread of IG and TimeSHAP, 0.061 and 0.068. The one ordering the floor supports is channel allocation in MATR: feature occlusion divides attribution among channels less accurately than IG and TimeSHAP, 0.377 against 0.294 and 0.301, a difference of 0.083 that exceeds the spread of every method, 0.039–0.054, and all ten occlusion models, 0.376–0.414, lie above every model of the other two, 0.287–0.343. Under uniform weighting, where the floor is measured for IG and occlusion only, the ordering holds narrowly: 0.364 against 0.292 for IG and 0.297 for TimeSHAP, a difference of 0.072 against spreads of 0.071 for IG and 0.057 for occlusion.

Mass on zero-weight channels is error by construction. It is zero on the reference model and, on the trained model, 0.134–0.198 in MATR, where two zero-weight channels, internal resistance and temperature, carry the degradation state (Table 4), and 0.037–0.040 in HUST and 0.039–0.048 in FORM, whose zero-weight channels are the null channels. No trained-model score of the graded field is void, since each primary model uses every weighted channel; the paired retrieval of FORM on the trained model is void, since the model is not shown to learn the pattern term (Section 4.2). TimeSHAP's values are stable under its sampling: across its three seeds its unit means differ by at most 0.003 in rank agreement and 0.02 positions in temporal profile error in every cell. Explaining only 40 of the 120 windows, chosen by a seeded rule, would give TimeSHAP values close to these: their means lie inside the 120-window intervals on both models, profiles and weightings, except in two HUST cells under uniform weighting, the trained-model temporal error, 3.56 against [3.03, 3.53], and the reference-model allocation error, 0.051 against [0.052, 0.080].

Two declared choices move the values. From the average-event background, scored against its own ground truth, TimeSHAP's rank agreement rises from 0.493 to 0.527 in MATR and falls from 0.739 to 0.697 in HUST and from 0.771 to 0.740 in FORM, and its temporal profile error rises from 1.82 to 2.21 positions in MATR, changes by 0.01 in HUST and falls from 3.82 to 3.57 in FORM. The background thus moves values by up to 0.04 in rank agreement and 0.4 positions, not in the same direction in every profile, so a reference value is quoted with its baseline. In rank agreement TimeSHAP stays above the other two methods in MATR, moves from between them to below both in HUST, and from first to last in FORM, by differences smaller than the spread of the floor. Under uniform weighting, on the models trained for that weighting, rank agreement changes by at most 0.04 in MATR and HUST, to 0.453–0.487 and 0.701–0.711 against ceilings of 0.856 and 0.930, but falls in FORM, to 0.532–0.570 against a ceiling of 0.826; the allocation error changes by at most 0.013 in every profile. Under uniform weighting the temporal profile error is 4.76–5.47 positions in MATR, 3.15–3.32 in HUST and 3.63–4.25 in FORM, against ceilings of 0.49, 0.24 and 0.82, above the 2.59, 1.70 and 1.77 positions that a fully permuted ground truth scores on the windows of Section 4.3: under uniform weighting, every method on a trained model places attribution along the window further from the ground truth than a random permutation of the ground truth does. The three methods keep their order under both weightings in every score in HUST, and in MATR except in the temporal profile error, where IG and TimeSHAP swap; in FORM their order changes in every score, by less than the spread of the floor.

Table 6: Reference values per attribution method, on the trained model and on the reference model, per profile, under recency weighting, on 120 windows of generated test units; every method explains every window. Rank agreement is measured against the graded field; the channel allocation error and the temporal profile error (in positions) are errors, zero for a perfect map; mass on zero-weight channels is error by construction. Unit means with 95% bootstrap intervals over test units; TimeSHAP rows use coalition-sampling seed 0. The exact attribution $w(x - x^0)$ of the reference model is the ceiling of its columns; the methods on the reference model, which equal it by construction, are shown as unit means only. The ensemble range is the smallest and largest unit mean over the ten equally accurate models of Section 3.6; a difference between methods on a trained model that is smaller than this range cannot be attributed to the method. The last row runs TimeSHAP from the average event it uses by convention, scored against the attribution ground truth for that baseline. Paired retrieval (average precision of the paired difference against the sparse set), FORM, on the 11 test units whose windows contain the sparse set: reference model IG 1.00; Occlusion 1.00; TimeSHAP 1.00; trained model void, since removing the pattern term from the training target does not change the model's error (0.0000 [−0.0001, 0.0002]).

	Rank agreement		Allocation error		Temporal error		Zero-weight mass	
Method	trained	reference	trained	reference	trained	reference	trained	reference
MATR (120 windows from 45 test units; every method on every window)								
IG	0.485 [0.445, 0.519]	0.869	0.294 [0.277, 0.314]	0.075	1.86 [1.68, 2.03]	0.38	0.134 [0.117, 0.153]	0.000
Occlusion	0.420 [0.368, 0.466]	0.869	0.377 [0.353, 0.407]	0.075	2.56 [2.39, 2.76]	0.38	0.198 [0.171, 0.233]	0.000
TimeSHAP	0.493 [0.454, 0.527]	0.870	0.301 [0.283, 0.321]	0.075	1.82 [1.65, 2.00]	0.38	0.136 [0.120, 0.155]	0.000
Exact $w(x - x^0)$	–	0.869	–	0.075	–	0.38	–	0.000
		[0.817, 0.909]		[0.060, 0.097]		[0.31, 0.45]		
Range, IG	0.406–0.506	–	0.287–0.341	–	1.86–2.73	–	–	–
Range, Occlusion	0.326–0.449	–	0.376–0.414	–	2.32–3.36	–	–	–
Range, TimeSHAP	0.415–0.514	–	0.292–0.343	–	1.82–2.55	–	–	–
TimeSHAP, average event	0.527 [0.488, 0.558]	–	0.313 [0.292, 0.341]	–	2.21 [2.07, 2.35]	–	0.138 [0.122, 0.166]	–
HUST (120 windows from 45 test units; every method on every window)								
IG	0.734 [0.700, 0.758]	0.957	0.164 [0.151, 0.179]	0.063	2.82 [2.65, 3.00]	0.18	0.037 [0.031, 0.051]	0.000
Occlusion	0.750 [0.715, 0.776]	0.957	0.182 [0.166, 0.203]	0.063	2.93 [2.77, 3.10]	0.18	0.040 [0.032, 0.057]	0.000
TimeSHAP	0.739 [0.705, 0.764]	0.957	0.176 [0.161, 0.191]	0.063	2.86 [2.71, 3.03]	0.18	0.038 [0.032, 0.052]	0.000
Exact $w(x - x^0)$	–	0.957	–	0.063	–	0.18	–	0.000
		[0.922, 0.978]		[0.051, 0.079]		[0.14, 0.24]		
Range, IG	0.597–0.747	–	0.125–0.218	–	2.63–3.80	–	–	–
Range, Occlusion	0.566–0.772	–	0.158–0.257	–	2.81–3.54	–	–	–
Range, TimeSHAP	0.600–0.781	–	0.136–0.230	–	2.61–3.81	–	–	–
TimeSHAP, average event	0.697 [0.664, 0.721]	–	0.156 [0.142, 0.174]	–	2.85 [2.71, 2.99]	–	0.030 [0.026, 0.041]	–
FORM (120 windows from 45 test units; every method on every window)								
IG	0.770 [0.723, 0.806]	0.849	0.215 [0.197, 0.240]	0.125	3.89 [3.72, 4.12]	0.63	0.039 [0.033, 0.045]	0.000
Occlusion	0.744 [0.696, 0.783]	0.849	0.265 [0.244, 0.289]	0.125	3.71 [3.49, 4.05]	0.63	0.048 [0.042, 0.055]	0.000
TimeSHAP	0.771 [0.725, 0.808]	0.850	0.220 [0.203, 0.245]	0.125	3.82 [3.65, 4.04]	0.63	0.039 [0.033, 0.045]	0.000
Exact $w(x - x^0)$	–	0.849	–	0.125	–	0.63	–	0.000
		[0.795, 0.891]		[0.103, 0.165]		[0.53, 0.79]		
Range, IG	0.438–0.776	–	0.196–0.257	–	2.97–3.89	–	–	–
Range, Occlusion	0.431–0.753	–	0.229–0.271	–	3.19–3.71	–	–	–
Range, TimeSHAP	0.458–0.779	–	0.194–0.262	–	2.82–3.82	–	–	–
TimeSHAP, average event	0.740 [0.696, 0.784]	–	0.247 [0.222, 0.273]	–	3.57 [3.35, 3.84]	–	0.047 [0.041, 0.054]	–

5 Discussion and future work

The research questions have different answers for the three profiles, and the differences concern what the benchmark can be used for rather than whether its construction holds. The construction holds in all three: the attribution ground truth sums to the decomposable target exactly, the reference model reproduces the target up to the weighted noise, the target is learnable, and every channel that enters it is used by every trained model. What varies is how much of each dataset a profile reproduces. In every profile, the spread of equally accurate trained models exceeds every difference between methods, with one exception in MATR.

What resists calibration. Generating per-unit offsets and correlated noise narrowed the transfer gap of MATR from 3.42 on the profile without them to 1.49 [1.04, 2.29], and moved HUST from 1.46 to 1.17 [0.94, 1.34] (Table 1); in both, the offsets account for the gain and the correlated noise for none of it. The gap of MATR is not closed: 1.49 lies outside the interval of HUST, the declared criterion, and the generator was not adjusted further. In Table 3, capacity alone transfers at 0.98 [0.92, 1.08], which points to the other channels, but the declared test does not locate the remainder there: the difference in log ratio, 0.42 [0.06, 0.90], does not hold when single seeds are left out. Removing the offsets from those channels no longer improves transfer, 1.93 [1.41, 2.58], as it did on the profile without offsets, so what remains is neither a constant level per unit nor a noise correlation shared by the units of a profile, both of which the generator now produces. FORM, fitted to cells of a different chemistry, format and laboratory, keeps a gap of the

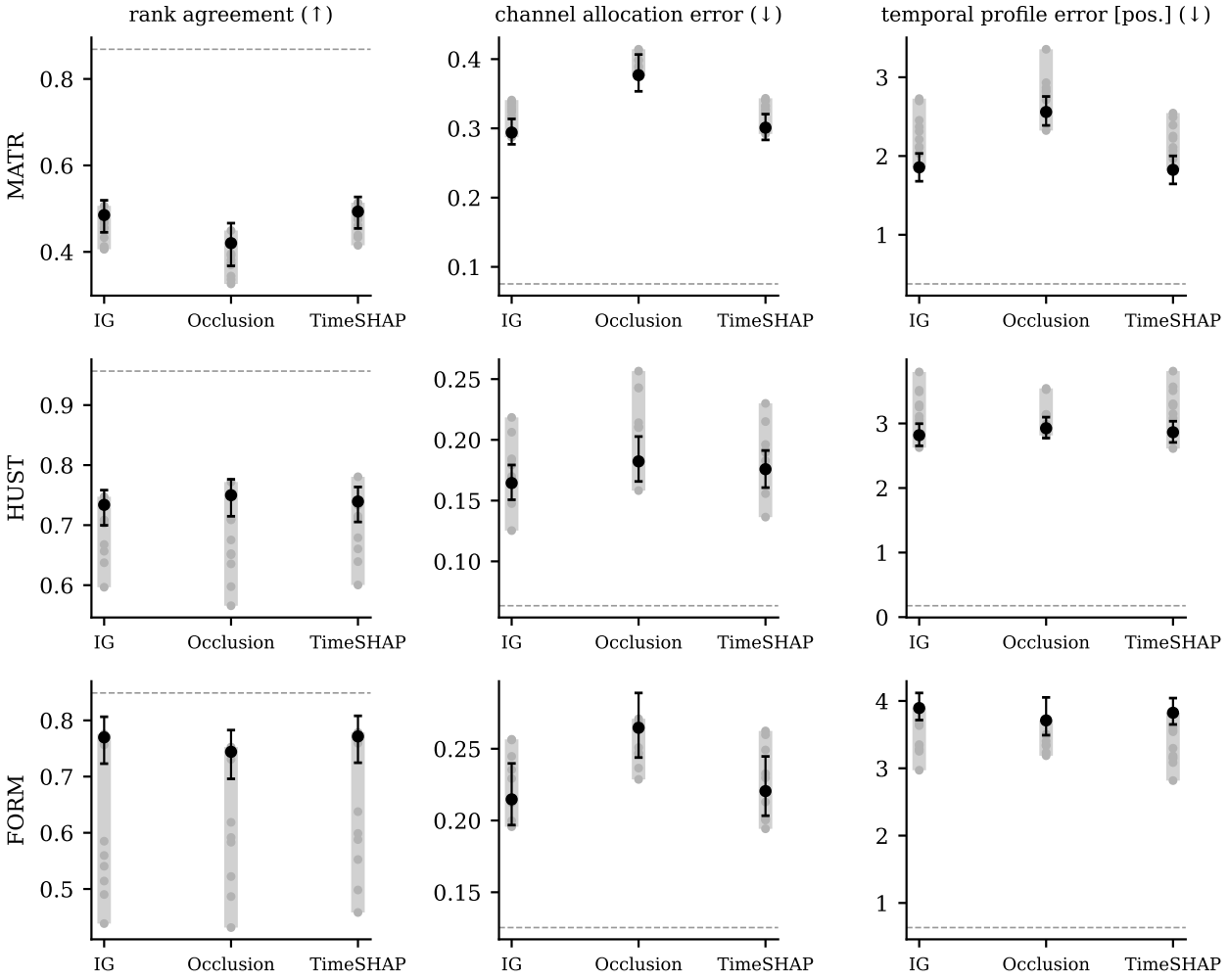


Figure 8: Trained-model scores against the identifiability floor, under recency weighting: for each method, the primary model's unit mean with its 95% interval (black) and the ten equally accurate ensemble members (grey points and band), with the exact attribution of the reference model as a dashed line.

same size, 1.54 [1.28, 1.74], and there the declared test does locate it: in the channels other than capacity, most of it in their levels early in a cell's life. The generator draws each cell's offsets together with its trajectory parameters, so a cell's level and its life stay paired, but an offset is the median over the whole life, and a mapping shared by all units cannot express a channel trajectory whose shape, and with it the early-life level relative to that median, differs between groups of cells. In MATR and FORM, the two profiles that keep a gap, the cells were charged or formed under protocols that differ from cell to cell, and in both the evidence points to the channels other than capacity; the cells of HUST share one charging protocol. Channel trajectories that differ in shape between protocols are therefore the evident, untested candidate. The covariance gap closed in all three profiles, in MATR and HUST through the offsets, and in none of the three through the correlated noise: sets drawn with independent noise reach about the same covariance agreement. HUST, whose interval contains one, therefore remains the profile on which the fidelity of the benchmark rests. A method evaluated on MATR or FORM has not been shown to behave the same way on a model that reads the structure the generator omits, and on FORM a discriminator tells generated from measured windows apart at an error of 0.10 to 0.17.

The sparse set. Of the evaluated profiles, only FORM carries a sparse set. Its capacity recovers after the diagnostic blocks that interrupt aging every 100 cycles, often enough for the detection rule to enable the type, at 2.28 times the rate in noise with the fitted variance and autocorrelation, and the recovery is attributed to the block rather than to rest alone, since after the 41 rests that leave the aging protocol unchanged the recoveries are not shown to cluster (2.04 [0, 4.51] times the rate at random positions). At its fitted rate and amplitude, however, the pattern term carries 1.7×10^{-5} of the

variance of the mean term, and a trained model is not shown to learn it (Table 4). The sparse set is therefore scored where the map is known to contain it. In the degraded-map test, paired retrieval falls from 1 to 0.37 under the smallest added noise and to about 0.1 when all mass is moved, but barely moves under smoothing (Section 4.3); on the reference model, the paired difference of every method's maps retrieves it exactly (Section 4.4). Whether a method recovers a localized effect that a trained model has learned is still not tested. That needs a dataset whose localized effects carry enough of the target for a trained model to use them; raising the amplitudes of FORM would supply one, but the profile would then no longer stand for its dataset. NASA PCoE, included for its regeneration, is not evaluated, and ISU-ILCC, which took over that role, does not fill it: on its fitting split the detection rule does not enable regeneration (positive capacity-residual runs at 1.66 times the rate in noise with the fitted variance and autocorrelation, against the declared factor of 2), a recovery after a reference test may reflect the kinetics of a partial discharge rather than regenerated capacity, and the profile is not taken through the protocol, since its offsets exceed the declared share of the target (Section 3.4).

What the scores register. In all three profiles, rank agreement against the graded field does not register two plausible failures of an attribution method, and registers mass shifted toward the start only when all of it is moved. Smoothing a correct map with a standard deviation of eight positions lowers rank agreement by at most 0.004, and moving any fraction of the mass short of the whole toward the end of the window lowers it by less than 0.002; under recency weighting the field grows smoothly toward the end, so neither operation reorders its cells (Table 5, Figure 7). The temporal profile error registers both, mass moved to the end at a fraction of 0.1 and smoothing at eight positions, the largest magnitude on the grid, while the channel allocation error registers what moves mass between channels and nothing positional. The two errors therefore separate what rank agreement conflates, mass on the wrong channels and mass in the wrong place along the window, and Table 6 reports both.

What the benchmark can adjudicate. The attribution ground truth is a property of the function that defines the decomposable target, not of the data alone. For the reference model that function is the model, so channel- and position-level ground truth is identifiable and a score measures the attribution method. For a trained model it is not: the channels share one degradation state, several functions of a window reproduce the target equally well, and a faithful attribution of the function a model learned can differ from $\phi^\star$. On the reference model the three methods return the exact attribution $w(x - x^0)$, TimeSHAP up to its sampling error, so that column checks correctness rather than ordering the methods. On a trained model the ten equally accurate models set the floor (Figure 8). In rank agreement they span 0.406 to 0.506 for Integrated Gradients in MATR, 0.597 to 0.747 in HUST and 0.438 to 0.776 in FORM, while the three methods differ by at most 0.073, 0.016 and 0.027, and neither the temporal profile error nor any score of HUST or FORM supports an ordering. One ordering survives the change of model, under both weightings: in MATR, feature occlusion divides attribution among the channels further from the ground truth, 0.377 against 0.294 and 0.301 in channel allocation error under recency weighting, a gap that exceeds the range of every method over the ten models; under uniform weighting the gap, 0.072, narrowly exceeds the ranges of Integrated Gradients and feature occlusion, the two methods on which the floor was measured. The per-unit offsets, which enter y and which the other channels do not fully predict, give a model reason to read the channels that carry them, and all ten models read every weighted channel (Table 4), yet the floor remains: using a channel does not fix how a model divides attribution among channels or along the window. In an exploratory pilot on HUST, a target defined on the observed window, $\sum w(x - x^0)$, narrowed every ensemble range of Integrated Gradients and occlusion by a factor of 1.6 to 15 and made the channel allocation identifiable by a small margin (ensemble ranges of 0.006 and 0.013 against a difference of 0.015 between the two methods), but not the position-level scores. The sparse set of FORM follows the same division: on the reference model every method localizes its patterns exactly, and on a trained model, which is not shown to learn them, the score is void. The reference model is therefore where the benchmark adjudicates; the trained columns say what a method returns on one model that fits the target, which is what a practitioner will have.

Limitations. Several declared choices bound the claims. The weighted channels have equal shares of the target, so nothing here shows how the scores behave when one channel dominates; the EOL threshold is fixed at 0.8 of nominal capacity; windows have 24 positions; the models are one architecture, and the ensemble that bounds identifiability varies initialization and size only; final-position weighting is not scored, and the scores have no measured operating range over generator settings. The within-unit correlation that sets the noise correlation is estimated from residuals about a degradation state read from noisy capacity, so capacity noise inflates the correlations of capacity with the other channels; the generator reproduces this bias and re-estimation adds it again, so the capacity pairs depart from the fitting split by 0.075 in generated units of MATR against 0.026 in held-out cells, and by 0.109 against 0.065 in FORM (Table 2). The discriminator cells of MATR and HUST in Table 1 are void under the readability rule; the shift of their seed means from the profiles without offsets, 0.07 to 0.33 in MATR and 0.16 to 0.49 in HUST, is the declared reading that the gap narrows, not a readable level. In FORM, 21 of the 300 generated units on which the properties are read end above the EOL threshold by more than the record-end tolerance and are left out of the property rows that depend on

EOL. No trained model is shown to learn the sparse set of FORM, so its retrieval scores rest on the reference model and on the degraded maps alone. Other rules were set after data were seen: the identifiability requirement in family selection; the use of units that never reach EOL where EOL is not needed; the treatment of records that end at the EOL threshold, also applied to generated series; the removal of readings that no degradation produces, and its extension in channels other than capacity to runs of up to three readings; the start of the EOL search at the early-life reference; the cap on pattern duration; the requirement that a pattern type occur a declared factor more often than the fitted noise produces it; the scoring of the sparse set on windows paired with their pattern-free counterparts; the minimum count and readability criterion of each measure; the probes that read the operating range of the profiles without offsets; the choice of HUST as the fidelity anchor, which the declared rule left open; and, for FORM, the reading of the charge current in the check that every cell is aged under one protocol, which as declared mistook the collapsing cycles at the end of life for a second protocol, and one sensitivity of the mechanism diagnostic, restricted to rests with no diagnostic block in the preceding eleven positions, which gives the same reading. The evidence that prompted each is released with the code. The declarations that govern the measurements on the profiles with offsets, including the generator changes, the two added scores, and the readings, were written after the results on the profiles without offsets were known and before those measurements were made, as was the choice of which measurements to drop. Those that govern FORM were written after every result on MATR, HUST and ISU-ILCC was known and before any file of the formation dataset was downloaded, but after its documentation had been read, which reports the initial capacities of its cells, the range of their cycle lives, and the finding that formation conditions drive cycle life, and after measured values of one of its cells, among them its capacity trajectory and the cycle at which it reached EOL, had been read; all that was read is listed with the declarations.

5.1 Future work

Learned component models. The generator fixes three components and a parametric family for each, which bounds the shapes a profile can express. A natural extension is to learn the components instead, with a model emitting a smooth, a localized, and an unstructured part that sum to the sequence. The attribution ground truth is unaffected, since it requires only that the generator record which part of each value it produced, not that the split be recovered from measurement; realism of x would improve while the answer key stays exact. The three roles would have to be retained, as the graded and sparse measures and the exclusion of noise are defined against them, and the present benchmark would serve to validate such a model, since it is where a known decomposition exists to recover.

Mappings per protocol. The remaining gaps of MATR and FORM point to channel trajectories whose shape differs between the protocols under which cells were charged or formed. A mapping per protocol group would let a profile express them, at the cost of levels set by the protocol, which enter the decomposable target through the weights and are bounded by the same share that stopped ISU-ILCC.

Observation-level target. The pilot above leaves open what would make positions, not only channels, identifiable on an observation-level target.

6 Conclusion

DEGRADX supplies attribution ground truth for a continuous target under sustained mean-nonstationarity, defined at every observation rather than at planted positions, and reports with each profile how closely it reproduces the measured dataset it was fitted to. Fitting it to three battery datasets shows what such a benchmark can claim. Fidelity is measurable and partial: per-unit levels carry the gains of the generator, correlated noise adds nothing measurable, on one profile a model is not distinguished in transfer from one trained on the measured data, and the two others keep gaps that point to channel trajectories differing in shape between the protocols under which cells were charged or formed. The ground truth is usable, since the target is learnable and every channel that enters it is read by trained models, but the one localized effect found in the data, a recovery of capacity after diagnostic blocks, carries too little of the target for a trained model to be shown to learn it, so the sparse part of the ground truth is checked on the reference model and on degraded maps alone. The choice of score matters: rank agreement, the score used against graded ground truth, misses smoothing and partial shifts of an attribution map, and the temporal profile error and the channel allocation error separate where along the window and on which channel a map is wrong. Finally, the reference model is where the benchmark adjudicates methods: on it the three methods recover the exact attribution, while on trained models the spread of equally accurate models exceeds the differences between methods in every score of every profile but channel allocation in MATR, so a trained-model score says what a method returns on one model that fits the target, not which method is more faithful. Learning the channel trajectories per protocol, finding degradation data whose localized effects carry part of the target, and making positions identifiable on trained models are the steps that remain.

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